Solutions to
Nonlinear Dynamics and Chaos (2nd ed.)

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Contents

3 Bifurcations **2**
3.1 Saddle-Node Bifurcation 2
3.2 Transcritical Bifurcation 5
3.3 Laser Threshold 7
3.4 Pitchfork Bifurcation 9
3.5 Overdamped Bead on a Rotating Hoop 19
3.6 Imperfect Bifurcations and Catastrophes 23
3.7 Insect Outbreak 28

Chapter 3

Bifurcations

3.1 Saddle-Node Bifurcation

Legend for all bifurcation diagrams: solid curves are stable fixed points, dashed curves are unstable fixed points.

Exercise 3.1.1 $\dot{x} = 1 + rx + x^2$.

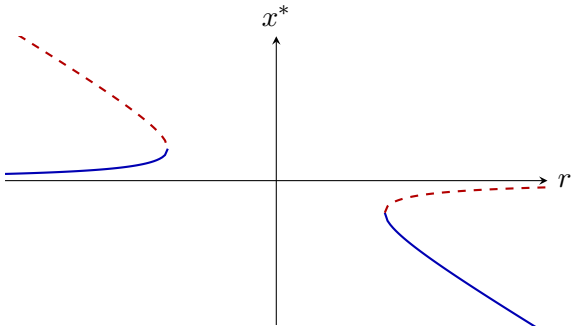
Solution. Fixed points solve $x^2 + rx + 1 = 0$:

$$x^*_\pm = \frac{-r \pm \sqrt{r^2 - 4}}{2}.$$

They exist only if $r^2 \geq 4$. Since $f'(x) = 2x + r$, we get $f'(x^*_\pm) = \pm\sqrt{r^2 - 4}$, so x^*_+ is unstable and x^*_- is stable whenever they exist.

Range of r	Fixed points	Flow
$r < -2$	two, both > 0: $x^*_- < 1 < x^*_+$	x^*_- stable, x^*_+ unstable
$r = -2$	one, $x^* = 1$	half-stable ($\dot{x} = (x-1)^2 \geq 0$)
$-2 < r < 2$	none	$\dot{x} > 0$ everywhere (flow to the right)
$r = 2$	one, $x^* = -1$	half-stable ($\dot{x} = (x+1)^2 \geq 0$)
$r > 2$	two, both < 0: $x^*_- < -1 < x^*_+$	x^*_- stable, x^*_+ unstable

Two saddle-node bifurcations occur, at $r = -2$ (where $x^* = 1$) and at $r = 2$ (where $x^* = -1$). At each, $\partial f/\partial r = x \neq 0$ and $\partial^2 f/\partial x^2 = 2 \neq 0$, so they are genuine saddle-nodes.

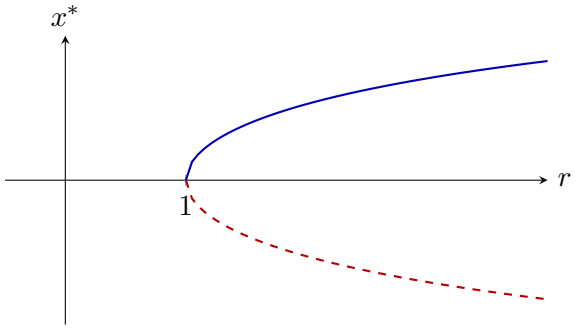


Exercise 3.1.2 $\dot{x} = r - \cosh x$.

Solution. Fixed points satisfy $\cosh x = r$. Since $\cosh x \geq 1$, they exist only for $r \geq 1$, and then $x^* = \pm \operatorname{arccosh} r = \pm \ln\bigl(r + \sqrt{r^2 - 1}\bigr)$. Here $f'(x) = -\sinh x$, so $x^* = +\operatorname{arccosh} r$ is stable and $x^* = -\operatorname{arccosh} r$ is unstable.

Range of r	Fixed points	Flow
$r < 1$	none	$\dot{x} < 0$ everywhere (flow to the left)
$r = 1$	$x^* = 0$	half-stable ($\dot{x} = 1 - \cosh x \le 0$)
$r > 1$	$x^* = \pm \operatorname{arccosh} r$	+ stable, − unstable

A saddle-node bifurcation occurs at $r_c = 1$, $x^* = 0$. Check: $f_r = 1 \neq 0$ and $f_{xx} = -\cosh 0 = -1 \neq 0$. Near the bifurcation, $\dot{x} \approx (r-1) - x^2/2$.



Exercise 3.1.3 $\dot{x} = r + x - \ln(1+x)$ (defined for $x > -1$).

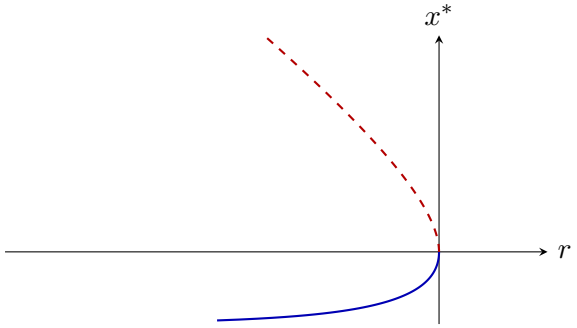
Solution. Here $f'(x) = 1 - \dfrac{1}{1+x} = \dfrac{x}{1+x}$, so f has its minimum on $(-1, \infty)$ at $x = 0$, with $f(0) = r$. Also $f \to +\infty$ as $x \to -1^+$ and as $x \to \infty$. Hence:

Range of r	Fixed points	Flow
$r > 0$	none	$\dot{x} > 0$ everywhere
$r = 0$	$x^* = 0$	half-stable ($\dot{x} = x - \ln(1+x) \ge 0$)
$r < 0$	$x_-^* \in (-1, 0)$ and $x_+^* > 0$	x_-^* stable, x_+^* unstable

(For $r < 0$, $f < 0$ between the two roots and $f > 0$ outside them, which gives the stability just stated.)

The saddle-node occurs at $r_c = 0$, $x^* = 0$, with $f_r = 1 \neq 0$ and $f_{xx}(0) = 1 \neq 0$. The normal form is $\dot{x} \approx r + \frac{1}{2}x^2$, so $x^* \approx \pm\sqrt{-2r}$ for small $r < 0$.

The fixed-point curve is $r = \ln(1+x) - x$, which we plot parametrically in x.



Exercise 3.1.4 $\dot{x} = r + \frac{1}{2}x - \dfrac{x}{1+x}$ (singular at $x = -1$).

Solution. Fixed points satisfy

$$r = R(x) := \frac{x}{1+x} - \frac{x}{2} = \frac{x(1-x)}{2(1+x)}.$$

Saddle-nodes occur where $R'(x) = 0$ (equivalently $f'(x) = \frac{1}{2} - \frac{1}{(1+x)^2} = 0$):

$$R'(x) = \frac{1 - 2x - x^2}{2(1+x)^2} = 0 \implies x^2 + 2x - 1 = 0 \implies x = -1 \pm \sqrt{2}.$$

The corresponding critical values are

$$x = \sqrt{2} - 1: \quad r_c = \tfrac{3}{2} - \sqrt{2} \approx 0.086, \qquad x = -1 - \sqrt{2}: \quad r_c = \tfrac{3}{2} + \sqrt{2} \approx 2.914.$$

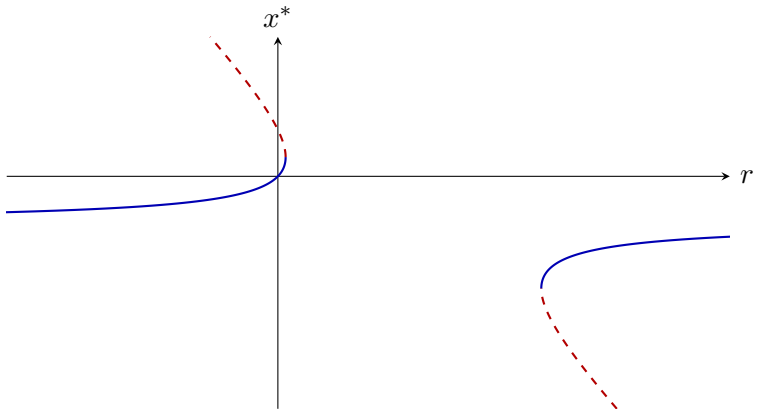
Shape of R. On $x > -1$: $R \to -\infty$ as $x \to -1^+$, R rises to a maximum $\frac{3}{2} - \sqrt{2}$ at $x = \sqrt{2} - 1$, then decreases to $-\infty$. On $x < -1$: $R \to +\infty$ as $x \to -1^-$ and as $x \to -\infty$, with a minimum $\frac{3}{2} + \sqrt{2}$ at $x = -1 - \sqrt{2}$.

Number of fixed points.

- $r < \frac{3}{2} - \sqrt{2}$: two fixed points in $x > -1$.
- $\frac{3}{2} - \sqrt{2} < r < \frac{3}{2} + \sqrt{2}$: none.
- $r > \frac{3}{2} + \sqrt{2}$: two fixed points in $x < -1$.

Stability. $f'(x) < 0$ iff $|1 + x| < \sqrt{2}$. So in each pair, the fixed point closer to $x = -1$ is stable and the farther one is unstable. Near the singularity, $f \to +\infty$ as $x \to -1^+$ and $f \to -\infty$ as $x \to -1^-$, so $x = -1$ is repelling from both sides (trajectories never cross it).

Both bifurcations are saddle-nodes (e.g. at $x = \sqrt{2} - 1$: $f_r = 1$, $f_{xx} = 2/(1+x)^3 \neq 0$). Sanity check at $r = 0$: $x^* = 0$ has $f' = -\frac{1}{2}$ (stable) and $x^* = 1$ has $f' = \frac{1}{4}$ (unstable).



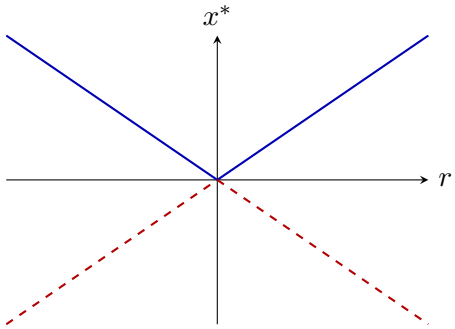
Exercise 3.1.5 (Unusual bifurcations) Sketch the vector fields and plot the fixed points as a function of r for (a) $\dot{x} = r^2 - x^2$, (b) $\dot{x} = r^2 + x^2$.

Solution. In both cases $a = \partial f/\partial r = 2r$ vanishes at the bifurcation point $(x, r) = (0, 0)$, so the standard saddle-node normal form does not apply.

(a) $\dot{x} = (r - x)(r + x)$, fixed points $x^* = r$ and $x^* = -r$. Since $f'(x) = -2x$:

- $r > 0$: $x^* = r$ is stable, $x^* = -r$ is unstable.
- $r < 0$: $x^* = r$ (negative) is unstable, $x^* = -r$ (positive) is stable.
- $r = 0$: $\dot{x} = -x^2$, a single half-stable fixed point.

So the stable branch is always $x^* = |r|$ and the unstable branch is $x^* = -|r|$. The two branches cross at the origin: there are two fixed points on both sides of $r = 0$, so none is created or destroyed. Only the labels (which line is stable) change. This is not a saddle-node.



(b) $\dot{x} = r^2 + x^2 \geq 0$. For $r \neq 0$ there are no fixed points and the flow is always to the right. At $r = 0$ there is exactly one fixed point $x^* = 0$, which is half-stable ($\dot{x} = x^2$). So the "bifurcation diagram" is the single isolated point $(r, x^*) = (0, 0)$: a fixed point exists for the single parameter value $r = 0$ and is born and annihilated instantly.

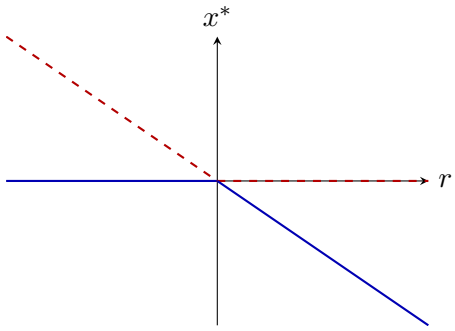
3.2 Transcritical Bifurcation

Exercise 3.2.1 $\dot{x} = rx + x^2$.

Solution. $\dot{x} = x(r + x)$, so $x^* = 0$ and $x^* = -r$. Since $f'(x) = r + 2x$: $f'(0) = r$ and $f'(-r) = -r$.

Range of r	Fixed points	Stability
$r < 0$	0 and $-r > 0$	0 stable, $-r$ unstable
$r = 0$	0	half-stable ($\dot{x} = x^2$)
$r > 0$	0 and $-r < 0$	0 unstable, $-r$ stable

A transcritical bifurcation occurs at $r_c = 0$: the two fixed points collide and exchange stability.



Exercise 3.2.2 $\dot{x} = rx - \ln(1 + x)$ (for $x > -1$).

Solution. $x^* = 0$ is a fixed point for every r. Here $f'(x) = r - \dfrac{1}{1+x}$, so $f'(0) = r - 1$: the origin is stable for $r < 1$ and unstable for $r > 1$.

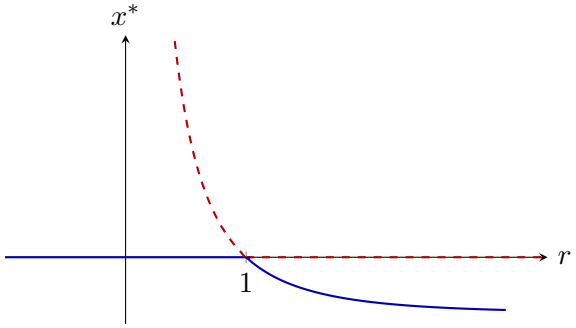
Other fixed points satisfy $r = \varphi(x) := \dfrac{\ln(1+x)}{x}$ for $x \neq 0$. The function φ is strictly decreasing on $(-1, \infty) \setminus \{0\}$, from $+\infty$ at $x \to -1^+$, through $\varphi \to 1$ as $x \to 0$, down to 0 as $x \to \infty$. Thus a nonzero fixed point $x^*(r)$ exists iff $r > 0$ (and $r \neq 1$), and it is unique:

$$0 < r < 1 : \ x^* > 0, \qquad r > 1 : \ -1 < x^* < 0.$$

Stability of $x^*(r)$: for $0 < r < 1$, $f(x) \to +\infty$ as $x \to \infty$, so $f > 0$ to the right of x^* and x^* is unstable; for $r > 1$, $f > 0$ to the left of x^* (as $f \to +\infty$ at $x \to -1^+$), and x^* is stable.

Range of r	Fixed points	Stability
$r \le 0$	0 only	stable
$0 < r < 1$	0 and $x^* > 0$	0 stable, x^* unstable
$r = 1$	0	half-stable ($\dot{x} \approx x^2/2$)
$r > 1$	0 and $-1 < x^* < 0$	0 unstable, x^* stable

A transcritical bifurcation occurs at $r_c = 1$ ($f_{xx} = 1 \neq 0$, $f_{xr} = 1 \neq 0$).



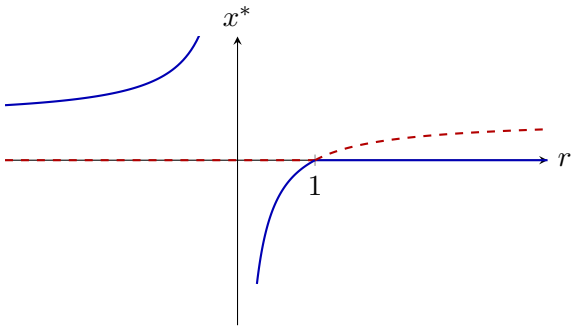
Exercise 3.2.3 $\dot{x} = x - rx(1-x)$.

Solution. $\dot{x} = x\big(1 - r + rx\big)$, so $x^* = 0$ and $x^* = \dfrac{r-1}{r} = 1 - \dfrac{1}{r}$ (for $r \neq 0$). We have $f'(x) = 1 - r + 2rx$, so

$$f'(0) = 1 - r, \qquad f'\Big(\tfrac{r-1}{r}\Big) = 1 - r + 2(r-1) = r - 1.$$

Range of r	Fixed points	Stability
$r < 0$	0 and $x^* > 1$	0 unstable, x^* stable
$r = 0$	0 only ($\dot{x} = x$)	unstable
$0 < r < 1$	0 and $x^* < 0$	0 unstable, x^* stable
$r = 1$	0 only ($\dot{x} = x^2$)	half-stable
$r > 1$	0 and $0 < x^* < 1$	0 stable, x^* unstable

A transcritical bifurcation occurs at $r_c = 1$. At $r = 0$ the second fixed point does not collide with $x = 0$; it runs off to $\pm\infty$, so this is not a bifurcation of the type studied here.



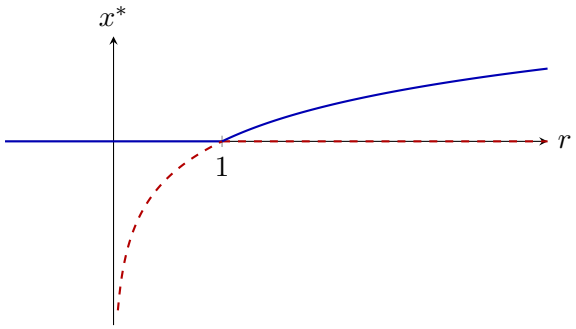
Exercise 3.2.4 $\dot{x} = x(r - e^x)$.

Solution. $x^* = 0$ always, and $e^x = r$ gives $x^* = \ln r$ when $r > 0$. With $f(x) = x(r - e^x)$,

$$f'(x) = (r - e^x) - xe^x, \qquad f'(0) = r - 1, \qquad f'(\ln r) = -r \ln r.$$

Range of r	Fixed points	Stability
$r \le 0$	0 only	stable
$0 < r < 1$	0 and $\ln r < 0$	0 stable, $\ln r$ unstable ($f' = -r \ln r > 0$)
$r = 1$	0	half-stable ($\dot{x} \approx -x^2$)
$r > 1$	0 and $\ln r > 0$	0 unstable, $\ln r$ stable ($f' < 0$)

A transcritical bifurcation occurs at $r_c = 1$ ($f_{xr} = 1 \neq 0$, $f_{xx}(0,1) = -2 \neq 0$).



Exercise 3.2.5 (Chemical kinetics) $A + X \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2X$, $\quad X + B \xrightarrow{k_2} C$, with a, b held constant.

Solution. **(a)** By the law of mass action, each step contributes to $\dot{x}$ as follows.

- $A + X \to 2X$: net gain of one X at rate $k_1 a x$.
- $2X \to A + X$: net loss of one X at rate $k_{-1} x^2$.
- $X + B \to C$: loss of one X at rate $k_2 b x$.

Therefore

$$\dot{x} = k_1 a x - k_{-1} x^2 - k_2 b x = c_1 x - c_2 x^2, \qquad c_1 = k_1 a - k_2 b, \quad c_2 = k_{-1}.$$

(b) Linearizing at $x^* = 0$ gives $\dot{x} \approx c_1 x$, so $x^* = 0$ is stable iff $c_1 < 0$, i.e. $k_2 b > k_1 a$. Chemically: the autocatalytic production of X (rate $k_1 a x$) is slower than its removal by B (rate $k_2 b x$), so any small amount of X is consumed and the concentration decays to zero. If $k_1 a > k_2 b$, X grows until the quadratic back-reaction balances it at $x^* = c_1/c_2$. The two regimes are separated by a transcritical bifurcation at $k_1 a = k_2 b$.

3.3 Laser Threshold

Exercise 3.3.1 (An improved model of a laser) $\dot{n} = GnN - kn$, $\quad \dot{N} = -GnN - fN + p$.

Solution. **(a)** Setting $\dot{N} \approx 0$: $N(Gn + f) = p$, so $N = \dfrac{p}{f + Gn}$. Substituting into the equation for n:

$$\dot{n} = n\left(\frac{Gp}{f + Gn} - k\right).$$

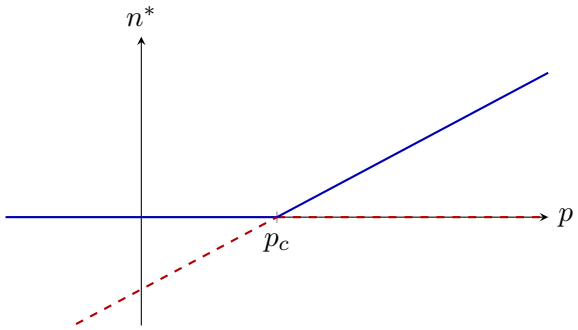
(b) $n^* = 0$ is a fixed point for every p. Linearizing, $\dot{n} \approx n\left(\frac{Gp}{f} - k\right)$, so $n^* = 0$ is unstable iff $\frac{Gp}{f} > k$, i.e.

$$p > p_c = \frac{kf}{G}.$$

(c) The second fixed point comes from $\dfrac{Gp}{f + Gn} = k$, giving

$$n^* = \frac{p}{k} - \frac{f}{G} = \frac{1}{k}\left(p - \frac{kf}{G}\right) = \frac{p - p_c}{k}.$$

It is positive exactly when $p > p_c$, and it passes through $n^* = 0$ at $p = p_c$ while exchanging stability with the origin. Hence a **transcritical bifurcation** occurs at the laser threshold $p = p_c$. For $p < p_c$ the laser acts as a lamp ($n^* = 0$); for $p > p_c$ it lases with $n^* = (p - p_c)/k$.



(The branch $n^* < 0$ for $p < p_c$ is unphysical, since $n \geq 0$.)

(d) The quasi-static approximation requires N to relax much faster than n evolves. For fixed n, the equation for N is linear, $\dot{N} = -(Gn + f)N + p$, so N relaxes at the rate $Gn + f$. Above threshold, the full system has fixed point $(n^*, N^*) = \left(\frac{p}{k} - \frac{f}{G},\ \frac{k}{G}\right)$ with Jacobian

$$J = \begin{pmatrix} GN^* - k & Gn^* \\ -GN^* & -(Gn^* + f) \end{pmatrix} = \begin{pmatrix} 0 & Gn^* \\ -k & -\lambda_N \end{pmatrix}, \qquad \lambda_N := Gn^* + f = \frac{Gp}{k}.$$

Its eigenvalues are $\lambda = \frac{1}{2}\big(- \lambda_N \pm \sqrt{\lambda_N^2 - 4kGn^*}\big)$. If $\lambda_N^2 \gg 4kGn^*$, the slow eigenvalue is

$$\lambda_{\text{slow}} \approx -\frac{kGn^*}{\lambda_N} = -k\Big(1 - \frac{p_c}{p}\Big),$$

which is exactly the relaxation rate of the reduced equation for n. So the approximation is valid when the decay rate of the atomic inversion is much larger than the photon decay rate:

$$\frac{Gp}{k} \gg k\Big(1 - \frac{p_c}{p}\Big) \quad \text{(above threshold)}, \qquad f \gg \Big|k - \frac{Gp}{f}\Big| \quad \text{(below threshold)}.$$

In particular it holds whenever $f \gg k$ and $Gp \gg k^2$ (the "fast N" regime), and it improves close to threshold, where the slow rate $\to 0$ (critical slowing down). One also needs the initial value of N to lie close to the slaved value $p/(f + Gn)$, or else there is a brief initial transient.

Exercise 3.3.2 (Maxwell–Bloch equations) $\dot{E} = \kappa(P - E)$, $\dot{P} = \gamma_1(ED - P)$, $\dot{D} = \gamma_2(\lambda + 1 - D - \lambda EP)$, with $\gamma_1, \gamma_2 \gg \kappa$.

Solution. **(a)** Set $\dot{P} \approx 0$ and $\dot{D} \approx 0$. From $\dot{P} = 0$: $P = ED$. Then $\dot{D} = 0$ gives

$$D = \lambda + 1 - \lambda EP = \lambda + 1 - \lambda E^2 D \implies D = \frac{\lambda + 1}{1 + \lambda E^2}, \qquad P = \frac{(\lambda + 1)E}{1 + \lambda E^2}.$$

Substituting into $\dot{E} = \kappa(P - E)$:

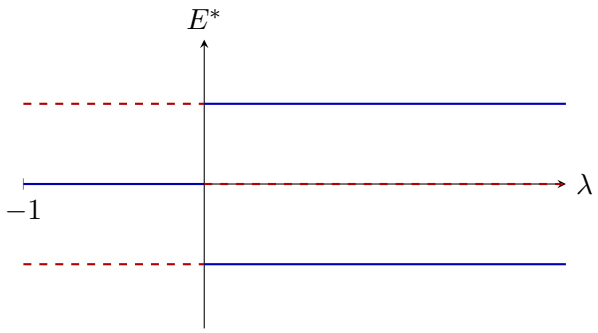
$$\dot{E} = \kappa E\left[\frac{\lambda + 1}{1 + \lambda E^2} - 1\right] = \frac{\kappa\lambda\, E\,(1 - E^2)}{1 + \lambda E^2}.$$

(b) Fixed points of the reduced equation: $E^* = 0$ and $E^* = \pm 1$. Both exist for every λ (with $\lambda \neq -1$ for $E^* = \pm 1$, since at $\lambda = -1$ the factor $1 - E^2$ cancels against the denominator and only $E^* = 0$ remains).

Stability. Write $g(E) = \kappa\lambda E(1 - E^2)/(1 + \lambda E^2)$.

- $E^* = 0$: $g'(0) = \kappa\lambda$. Stable for $\lambda < 0$, unstable for $\lambda > 0$.
- $E^* = \pm 1$: the factor $(1 - E^2)$ vanishes there, so only its derivative contributes: $g'(\pm 1) = \left.\frac{\kappa\lambda(1-3E^2)}{1+\lambda E^2}\right|_{E^2=1} = -\frac{2\kappa\lambda}{1+\lambda}$. For $\lambda > 0$ this is negative (stable); for $-1 < \lambda < 0$ it is positive (unstable).

(c) Bifurcation diagram (for $\lambda > -1$; solid = stable, dashed = unstable):



Notes:

- The three branches $E^* = 0, \pm 1$ do not meet, so this is not a standard pitchfork. At $\lambda = 0$ the reduced equation degenerates to $\dot{E} \equiv 0$: every E is a fixed point. For λ just above 0 the laser switches on, going from $E^* = 0$ to the two nonzero states $E^* = \pm 1$.
- The reduced equation is singular when $1 + \lambda E^2 = 0$, which happens for $E^2 = -1/\lambda$ if $\lambda < 0$. In that region the adiabatic elimination breaks down (the slaved value of D blows up), so the reduced equation should not be trusted there. For $\lambda < -1$ this singularity lies between 0 and ± 1, and the formula's prediction that $E^* = \pm 1$ is stable should be treated with caution.

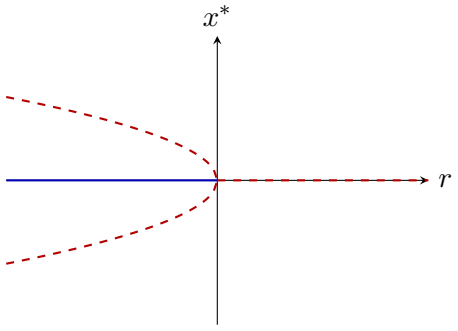
3.4 Pitchfork Bifurcation

Exercise 3.4.1 $\dot{x} = rx + 4x^3$.

Solution. $\dot{x} = x(r + 4x^2)$. Fixed points: $x^* = 0$, and $x^* = \pm\frac{1}{2}\sqrt{-r}$ for $r < 0$. We have $f'(x) = r + 12x^2$, so $f'(0) = r$ and, at $x^2 = -r/4$, $f' = r - 3r = -2r > 0$ (unstable).

Range of r	Fixed points	Stability
$r < 0$	$0,\ \pm\frac{1}{2}\sqrt{-r}$	0 stable; $\pm\frac{1}{2}\sqrt{-r}$ unstable
$r = 0$	0	unstable ($\dot{x} = 4x^3$)
$r > 0$	0	unstable

The nonzero fixed points exist for $r < 0$ and are unstable: a **subcritical pitchfork** at $r_c = 0$.



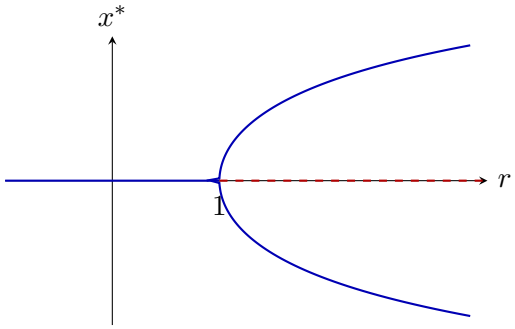
Exercise 3.4.2 $\dot{x} = rx - \sinh x$.

Solution. Fixed points: $x^* = 0$, and $\dfrac{\sinh x}{x} = r$ for $x \neq 0$. The function $\sinh x/x$ is even, equals 1 at $x \to 0$, and increases strictly in $|x|$ to ∞. So nonzero fixed points $\pm x^*(r)$ exist iff $r > 1$.

Stability: $f'(x) = r - \cosh x$. At 0: $f'(0) = r - 1$. At a nonzero fixed point ($r = \sinh x/x$): $f' = \dfrac{\sinh x - x\cosh x}{x}$. Since $\frac{\mathrm{d}}{\mathrm{d}x}(\sinh x - x\cosh x) = -x\sinh x < 0$ for $x \neq 0$, the numerator is negative for $x > 0$, so $f' < 0$ (and by symmetry also for $x < 0$): the nonzero fixed points are stable.

Range of r	Fixed points	Stability
$r < 1$	0	stable
$r = 1$	0	stable (weakly; $\dot{x} \approx -x^3/6$)
$r > 1$	$0,\ \pm x^*$	0 unstable; $\pm x^*$ stable

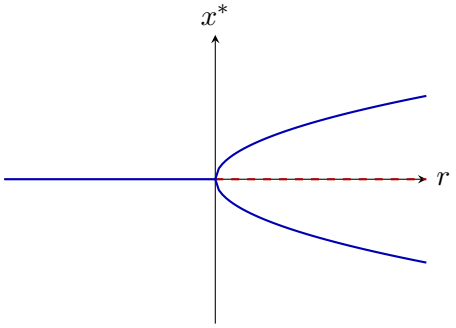
A **supercritical pitchfork** at $r_c = 1$. Near it, $\dot{x} \approx (r-1)x - \frac{1}{6}x^3$, so $x^* \approx \pm\sqrt{6(r-1)}$.



Exercise 3.4.3 $\dot{x} = rx - 4x^3$.

Solution. $\dot{x} = x(r - 4x^2)$. Fixed points: $x^* = 0$, and $x^* = \pm\frac{1}{2}\sqrt{r}$ for $r > 0$. With $f'(x) = r - 12x^2$: $f'(0) = r$, and at $x^2 = r/4$, $f' = r - 3r = -2r < 0$ (stable).

For $r < 0$ only $x^* = 0$, stable. For $r > 0$: 0 unstable, $\pm\frac{1}{2}\sqrt{r}$ stable. This is a **supercritical pitchfork** at $r_c = 0$.



Exercise 3.4.4 $\dot{x} = x + \dfrac{rx}{1+x^2}$.

Solution. $\dot{x} = x\left(1 + \dfrac{r}{1+x^2}\right)$. Fixed points: $x^* = 0$, and $1 + x^2 + r = 0$, i.e. $x^* = \pm\sqrt{-1-r}$ for $r < -1$.

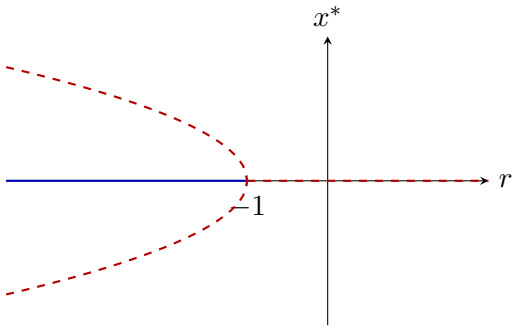
Stability: $f'(0) = 1 + r$. For the nonzero fixed points, put $1 + x^2 = -r$; then

$$f'(x) = 1 + \frac{r}{1+x^2} - \frac{2rx^2}{(1+x^2)^2} = 1 - 1 - \frac{2x^2}{r} = \frac{2(1+r)}{r} > 0 \quad (r < -1, \text{ since both } 1+r < 0 \text{ and } r < 0),$$

so they are unstable.

Range of r	Fixed points	Stability
$r < -1$	$0,\ \pm\sqrt{-1-r}$	0 stable; $\pm\sqrt{-1-r}$ unstable
$r = -1$	0	unstable ($\dot{x} \approx x^3$)
$r > -1$	0	unstable

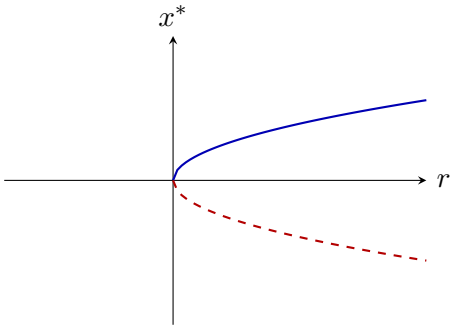
A **subcritical pitchfork** at $r_c = -1$.



Exercises 3.4.5–3.4.10 (identify the type of each bifurcation).

Exercise 3.4.5 $\dot{x} = r - 3x^2$.

Solution. Fixed points $x^* = \pm\sqrt{r/3}$ for $r > 0$; none for $r < 0$. $f'(x) = -6x$: $+\sqrt{r/3}$ is stable, $-\sqrt{r/3}$ is unstable. At $r = 0$, $\dot{x} = -3x^2$ (half-stable). A **saddle-node** bifurcation at $r_c = 0$.



Exercise 3.4.6 $\dot{x} = rx - \dfrac{x}{1+x}$ (singular at $x = -1$).

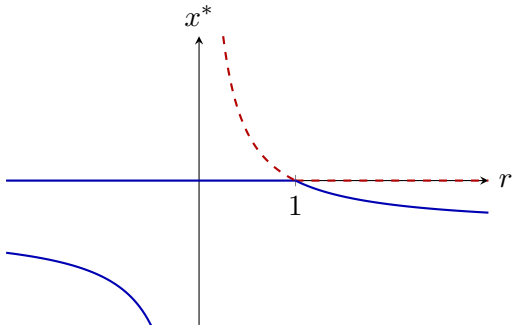
Solution. $\dot{x} = x\left(r - \dfrac{1}{1+x}\right)$. Fixed points: $x^* = 0$ and $x^* = \dfrac{1}{r} - 1$ (for $r \neq 0$). Here $f'(x) = r - \dfrac{1}{(1+x)^2}$, so

$$f'(0) = r - 1, \qquad f'\left(\tfrac{1}{r} - 1\right) = r - r^2 = r(1-r).$$

The two fixed points collide at $x = 0$ when $r = 1$. For $r = 1$, $\dot{x} = \dfrac{x^2}{1+x}$ (half-stable).

Range of r	Fixed points	Stability
$r < 0$	0 and $x^* < -1$	both stable
$0 < r < 1$	0 and $x^* > 0$	0 stable, x^* unstable
$r = 1$	0	half-stable
$r > 1$	0 and $-1 < x^* < 0$	0 unstable, x^* stable

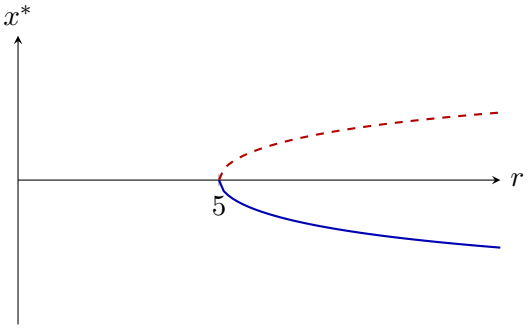
A **transcritical** bifurcation at $r_c = 1$. (At $r = 0$ the second fixed point runs off to $-\infty$; this is not a bifurcation of the local type. For $r < 0$ the second fixed point lies on the far side of the singularity at $x = -1$, so it is dynamically disconnected from $x = 0$.)



Exercise 3.4.7 $\dot{x} = 5 - re^{-x^2}$.

Solution. Fixed points satisfy $e^{-x^2} = 5/r$, so $x^2 = \ln(r/5)$: they exist iff $r \geq 5$. If $r < 5$, then $re^{-x^2} \leq \max(r, 0) < 5$ and $\dot{x} > 0$ everywhere. Also $f'(x) = 2rxe^{-x^2}$, which at a fixed point equals $10x$. Hence $x^* = -\sqrt{\ln(r/5)}$ is stable and $x^* = +\sqrt{\ln(r/5)}$ is unstable. At $r = 5$: $\dot{x} = 5(1 - e^{-x^2}) \geq 0$ (half-stable).

A **saddle-node** bifurcation at $r_c = 5$, $x^* = 0$.



Exercise 3.4.8 $\dot{x} = rx - \dfrac{x}{1+x^2}$.

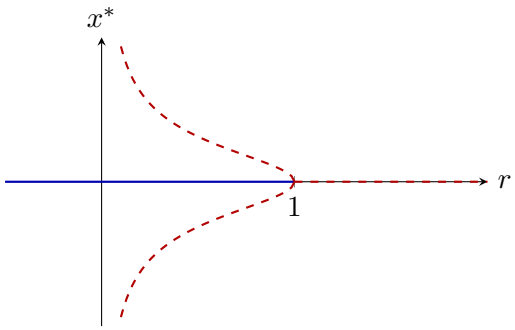
Solution. $\dot{x} = x\left(r - \dfrac{1}{1+x^2}\right)$. Fixed points: $x^* = 0$, and $1 + x^2 = 1/r$, i.e. $x^* = \pm\sqrt{\frac{1}{r} - 1}$ for $0 < r < 1$. $f'(0) = r - 1$. For the nonzero fixed points, set $u = 1 + x^2 = 1/r$:

$$f' = r - \frac{1 - x^2}{(1+x^2)^2} = \frac{1}{u} - \frac{2 - u}{u^2} = \frac{2(u-1)}{u^2} = \frac{2x^2}{u^2} > 0,$$

so they are unstable.

Range of r	Fixed points	Stability
$r \le 0$	0	stable
$0 < r < 1$	$0,\ \pm\sqrt{1/r - 1}$	0 stable; $\pm\sqrt{1/r-1}$ unstable
$r = 1$	0	weakly unstable ($\dot{x} \approx x^3$)
$r > 1$	0	unstable

A **subcritical pitchfork** at $r_c = 1$. (As $r \to 0^+$ the unstable branches run off to infinity; $r = 0$ is not a bifurcation.)



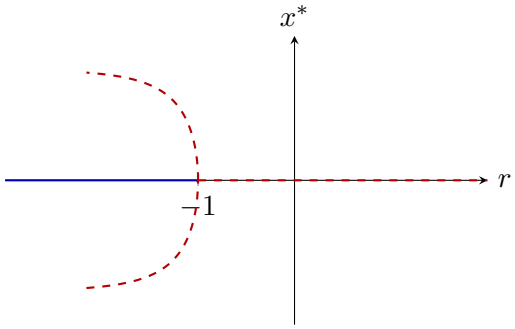
Exercise 3.4.9 $\dot{x} = x + \tanh(rx)$.

Solution. $x^* = 0$ is always a fixed point; $f'(0) = 1 + r$, so 0 is stable for $r < -1$ and unstable for $r > -1$. Nonzero fixed points solve $\tanh(rx) = -x$. For $r \ge 0$ the left side has the same sign as x, so there are none. For $r < 0$, $y = \tanh(rx)$ is decreasing with slope r at the origin and saturates at ∓ 1, so it meets the line $y = -x$ at a nonzero point exactly when its slope at 0 is steeper than -1, i.e. $r < -1$. Then the solutions are $\pm x^*$ with $0 < x^* < 1$, given parametrically by

$$r = -\frac{\operatorname{artanh} x}{x}.$$

Stability: for $r < -1$, $f(x) \approx (1 + r)x < 0$ for small $x > 0$, while $f(x) > 0$ for $x \ge 1$; so f crosses from negative to positive at x^*, and x^* is unstable (and likewise $-x^*$). Near $r = -1$: $\dot{x} \approx (1 + r)x + \frac{1}{3}x^3$ (using $-r^3 \approx 1$), so $x^* \approx \pm\sqrt{-3(1 + r)}$.

A **subcritical pitchfork** at $r_c = -1$: 0 is stable for $r < -1$ with two unstable nonzero fixed points; for $r > -1$ only an unstable origin remains.



Exercise 3.4.10 $\dot{x} = rx + \dfrac{x^3}{1 + x^2}$.

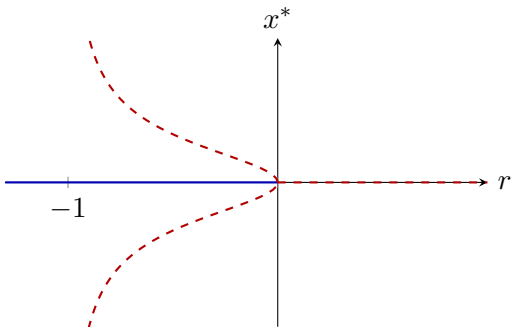
Solution. $\dot{x} = x\left(r + \dfrac{x^2}{1 + x^2}\right)$. Fixed points: $x^* = 0$, and $\dfrac{x^2}{1 + x^2} = -r$, i.e. $x^* = \pm\sqrt{\dfrac{-r}{1 + r}}$, which exist for $-1 < r < 0$. $f'(0) = r$. At a nonzero fixed point (with $r = -x^2/(1 + x^2)$):

$$f' = r + \frac{3x^2 + x^4}{(1 + x^2)^2} = \frac{-x^2(1 + x^2) + 3x^2 + x^4}{(1 + x^2)^2} = \frac{2x^2}{(1 + x^2)^2} > 0,$$

so they are unstable.

Range of r	Fixed points	Stability
$r \le -1$	0	stable
$-1 < r < 0$	$0,\ \pm\sqrt{-r/(1+r)}$	0 stable; $\pm\sqrt{-r/(1+r)}$ unstable
$r = 0$	0	unstable ($\dot{x} \approx x^3$)
$r > 0$	0	unstable

A **subcritical pitchfork** at $r_c = 0$. (As $r \to -1^+$ the unstable branches run off to infinity; $r = -1$ is not a bifurcation.)



Exercise 3.4.11 (An interesting bifurcation diagram) $\dot{x} = rx - \sin x$.

Solution. **(a)** For $r = 0$: $\dot{x} = -\sin x$, with fixed points $x^* = n\pi$ for all integers n. Since $f'(x) = -\cos x$, the points $x^* = 2k\pi$ are stable ($f' = -1$) and $x^* = (2k+1)\pi$ are unstable ($f' = +1$). The flow runs toward each even multiple of π from both sides, away from each odd multiple.

(b) Let $r > 1$ and $x \neq 0$. Then $|\sin x| \le |x| < r|x|$, so $rx - \sin x$ has the sign of x. Hence $x^* = 0$ is the *only* fixed point; $f'(0) = r - 1 > 0$, so it is unstable (a source): all trajectories flow away from it to $\pm\infty$.

(c) Nonzero fixed points satisfy $r = \dfrac{\sin x}{x} =: \rho(x)$. At $r = 1$ the origin changes stability. Expanding, $\dot{x} \approx (r-1)x + \frac{1}{6}x^3$, so as r decreases through 1 a pair of nonzero fixed points $\pm\sqrt{6(1-r)}$ is born, and they are *unstable*: a **subcritical pitchfork** at $r = 1$.

As r decreases further, new fixed-point pairs appear whenever the horizontal line at height r becomes tangent to a local *maximum* of $\rho(x)$. Extrema of ρ satisfy $\rho' = 0 \iff \tan x = x$, and there $\rho = \cos x$. The positive solutions are $x_1 \approx 4.493$, $x_2 \approx 7.725$, $x_3 \approx 10.904$, $x_4 \approx 14.066, \ldots$ with $\rho(x_n) \approx -0.2172,\ 0.1284,\ -0.0913,\ 0.0709, \ldots$ (even n are maxima, odd n are minima). So for r decreasing from ∞ to 0 there is an infinite sequence of **saddle-node** bifurcations (each occurring symmetrically at $\pm x_n$, creating a stable/unstable pair) at

$$r \approx 0.1284\ (x \approx \pm 7.725), \quad 0.0709\ (x \approx \pm 14.066), \quad \ldots$$

accumulating at $r = 0^+$.

(d) For $0 < r \ll 1$ the relevant extrema lie at large x. The solutions of $\tan x = x$ are then close to $x \approx (n + \frac{1}{2})\pi$, and at the maxima $x \approx 2k\pi + \frac{\pi}{2}$ we have $\sin x \approx 1$, so

$$r_k \approx \frac{1}{x} \approx \frac{1}{2\pi k + \pi/2} = \frac{2}{(4k+1)\pi}, \qquad k = 1, 2, 3, \ldots$$

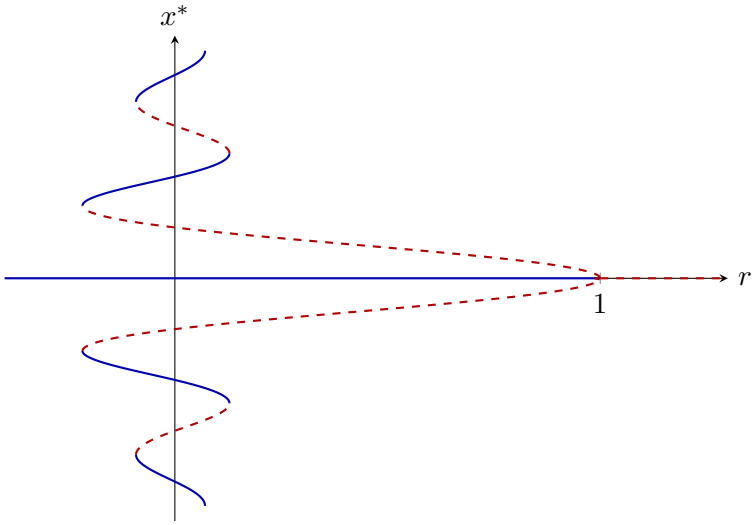
(e.g. $k = 1$: 0.127, compared with the exact 0.128).

(e) For r decreasing from 0 to $-\infty$ the origin stays stable ($f'(0) = r - 1 < 0$), so there is no further pitchfork. Fixed pairs are *destroyed* in saddle-node bifurcations whenever the line at height r passes a local minimum of ρ: at the odd x_n, i.e. near $x \approx 2\pi k + \frac{3\pi}{2}$, giving

$$r \approx -\frac{2}{(4k+3)\pi} \quad (k = 0, 1, 2, \dots): \quad r \approx -0.2172\ (x_1),\ -0.0913\ (x_3),\ -0.0580\ (x_5),\ \dots$$

These are again infinitely many saddle-node bifurcations, now accumulating at $r = 0^-$ and ordered so that the *largest* $|x|$ goes first as r decreases from 0. The last one is at $r \approx -0.2172$, $x \approx \pm 4.493$. For $r < -0.2172$ the only fixed point is the stable origin.

(f) Stability of a nonzero fixed point: $f' = r - \cos x = \dfrac{\sin x - x\cos x}{x}$, and $h(x) = \sin x - x\cos x$ satisfies $h(0) = 0$, $h' = x\sin x$. So for $x > 0$: $f' > 0$ (unstable) on $(0, x_1)$; $f' < 0$ (stable) on (x_1, x_2); unstable on (x_2, x_3); stable on (x_3, x_4); and so on, alternating at each x_n. The picture is symmetric under $x \to -x$. Below, solid = stable, dashed = unstable (plotted for $|x| \le 14$).

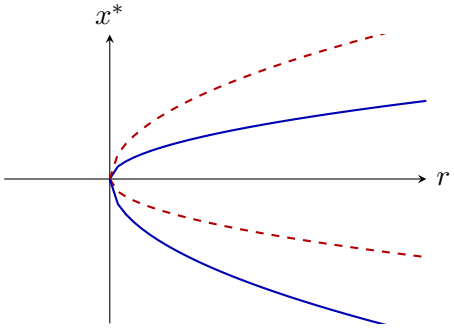


Exercise 3.4.12 ("Quadfurcation") Find $\dot{x} = f(x, r)$ with no fixed points for $r < 0$ and four branches of fixed points for $r > 0$; generalize.

Solution. Take

$$\dot{x} = (r - x^2)(4r - x^2).$$

For $r < 0$ both factors are positive, so $\dot{x} > 0$ and there are no fixed points. For $r > 0$ the fixed points are $x^* = \pm\sqrt{r}$ and $x^* = \pm 2\sqrt{r}$: four branches emerging from the origin at $r = 0$ (where $\dot{x} = x^4$). Checking signs of $\dot{x}$: $\dot{x} > 0$ for $x > 2\sqrt{r}$, < 0 on $(\sqrt{r}, 2\sqrt{r})$, > 0 on $(-\sqrt{r}, \sqrt{r})$, < 0 on $(-2\sqrt{r}, -\sqrt{r})$, > 0 for $x < -2\sqrt{r}$. So $\pm\sqrt{r}$: $+\sqrt{r}$ stable, $-\sqrt{r}$ unstable; $+2\sqrt{r}$ unstable, $-2\sqrt{r}$ stable.



General case. For $2n$ branches use

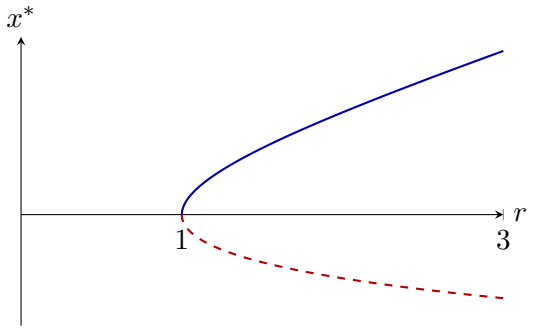
$$\dot{x} = \prod_{k=1}^{n} \left(k^2 r - x^2\right),$$

which has no fixed points for $r < 0$ and the $2n$ fixed points $\pm k\sqrt{r}$ $(k = 1, \ldots, n)$ for $r > 0$. For an odd number $2n+1$ of branches use $\dot{x} = x\prod_{k=1}^{n}(k^2 r - x^2)$: this has the single fixed point $x = 0$ for $r < 0$ and $2n+1$ fixed points for $r > 0$ (for $n = 1$ this is $x(r - x^2)$, the supercritical pitchfork, a "trifurcation"). (Heuristically, if f has the same sign at $x \to \pm\infty$ and no fixed points for $r < 0$, then for $r > 0$ it has an even number of sign-changing roots, so an odd number of branches requires an extra fixed point such as $x = 0$ that is present on both sides of $r = 0$.)

Exercise 3.4.13 (Computer work) Plot x^* vs. r for $0 \le r \le 3$ for (a) $\dot{x} = r - x - e^{-x}$, (b) $\dot{x} = 1 - x - e^{-rx}$.

Solution. **Easy way: solve for r in terms of x** and plot the parametric curve $(r(x), x)$.

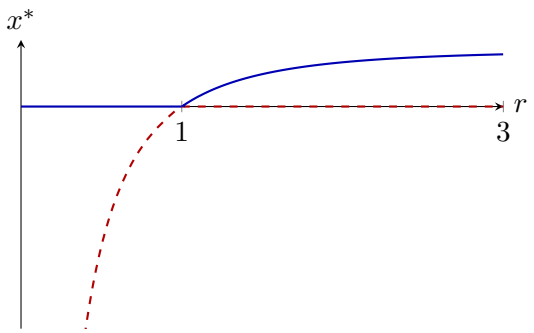
(a) $\dot{x} = 0 \iff r = x + e^{-x} =: g(x)$. Since $g'(x) = 1 - e^{-x}$, g has its minimum $g(0) = 1$ at $x = 0$ and $g \to \infty$ as $x \to \pm\infty$. So there are no fixed points for $r < 1$, one $(x^* = 0)$ at $r = 1$, and two for $r > 1$, one on each side of 0. With $f'(x) = -1 + e^{-x}$ the branch $x > 0$ is stable and $x < 0$ is unstable: a **saddle-node** at $r = 1$. At $r = 3$ the branches end at $x^* \approx 2.948$ (stable) and $x^* \approx -1.505$ (unstable).



(b) $x^* = 0$ is a fixed point for all r. Other fixed points satisfy $e^{-rx} = 1 - x$, i.e.

$$r = \psi(x) := -\frac{\ln(1-x)}{x}, \qquad x < 1,\ x \neq 0.$$

$\psi(x) = 1 + \frac{x}{2} + \frac{x^2}{3} + \cdots$ is strictly increasing, with $\psi \to 1$ as $x \to 0$, $\psi \to \infty$ as $x \to 1^-$, and $\psi \to 0^+$ as $x \to -\infty$. So for each $r > 0$, $r \neq 1$, there is exactly one nonzero fixed point: $x^* \in (0, 1)$ if $r > 1$ and $x^* < 0$ if $0 < r < 1$. Since $f'(0) = r - 1$, the origin is stable for $r < 1$ and unstable for $r > 1$, and the other branch has the opposite stability: a **transcritical** bifurcation at $r = 1$. At $r = 3$: $x^* \approx 0.940$.



Harder way (Newton–Raphson). For each r on a grid $0 = r_0 < r_1 < \cdots < r_M = 3$, solve $f(x, r) = 0$ by iterating

$$x_{n+1} = x_n - \frac{f(x_n, r)}{f_x(x_n, r)},$$

using the converged solution at r_{j-1} as the initial guess for r_j (continuation). For (a): $f = r - x - e^{-x}$, $f_x = -1 + e^{-x}$; for (b): $f = 1 - x - e^{-rx}$, $f_x = -1 + re^{-rx}$. Near the bifurcation point $f_x \to 0$, so Newton's method converges slowly and needs a small step in r or a start on the correct side of $x = 0$; to follow each branch, start from a guess on that branch, e.g. $x_0 > 0$ and $x_0 < 0$ in (a).

Exercise 3.4.14 (Subcritical pitchfork) $\dot{x} = rx + x^3 - x^5$.

Solution. **(a)** $\dot{x} = x\,(r + x^2 - x^4)$. Fixed points: $x^* = 0$ and $x^4 - x^2 - r = 0$, i.e.

$$x^2 = \frac{1 \pm \sqrt{1 + 4r}}{2} =: s_\pm.$$

Real solutions need $1 + 4r \geq 0$. $s_+ > 0$ for all $r > -\frac{1}{4}$; $s_- \geq 0$ iff $\sqrt{1+4r} \leq 1$ iff $r \leq 0$. Hence:

- $r < -\frac{1}{4}$: only $x^* = 0$;
- $-\frac{1}{4} < r < 0$: five fixed points, 0 and $\pm\sqrt{s_+}$, $\pm\sqrt{s_-}$;
- $r > 0$: three fixed points, 0 and $\pm\sqrt{s_+}$.

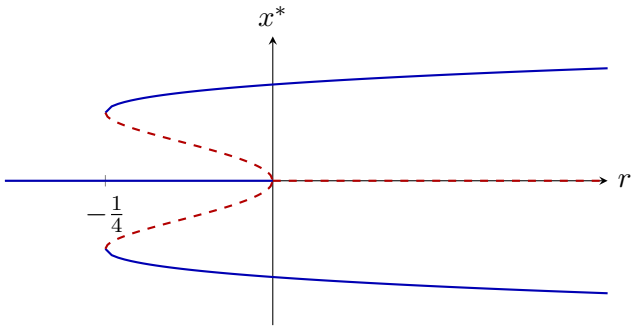
(b) $f'(x) = r + 3x^2 - 5x^4$. At 0: $f' = r$. At a nonzero fixed point, using $x^4 = x^2 + r$: $f' = r + 3x^2 - 5x^2 - 5r = -4r - 2x^2$, so with $x^2 = s_\pm$,

$$f' = -\sqrt{1 + 4r}\,\big(\sqrt{1 + 4r} \pm 1\big).$$

For s_+ this is negative: *stable*. For s_- (with $-\frac{1}{4} < r < 0$, so $\sqrt{1+4r} < 1$) it is positive: *unstable*. Summary:

Range of r	Stable	Unstable
$r < -\frac{1}{4}$	0	—
$-\frac{1}{4} < r < 0$	0, $\pm\sqrt{s_+}$	$\pm\sqrt{s_-}$
$r > 0$	$\pm\sqrt{s_+}$	0

(c) The nonzero large-amplitude fixed points are born at $r_s = -\frac{1}{4}$, where $s_+ = s_- = \frac{1}{2}$, i.e. at $x^* = \pm\frac{1}{\sqrt{2}}$, in a pair of saddle-node bifurcations. (The subcritical pitchfork itself, with the small unstable branches $\pm\sqrt{s_-}$ merging into 0, is at $r = 0$.)



Exercise 3.4.15 (First-order phase transition) For $\dot{x} = rx + x^3 - x^5$, find r_c at which the potential V has three equally deep wells.

Solution. With $\dot{x} = -V'(x)$,

$$V(x) = -\frac{r}{2}x^2 - \frac{1}{4}x^4 + \frac{1}{6}x^6, \qquad V(0) = 0.$$

The wells at 0 and $\pm\sqrt{s_+}$ are equally deep iff $V(\pm\sqrt{s_+}) = 0$. Write $s = x^2$ at the large fixed point, so $s^2 = s + r$. Then

$$V = -\frac{r}{2}s - \frac{1}{4}s^2 + \frac{1}{6}s^3 = s\Big(-\frac{r}{2} - \frac{s}{4} + \frac{s^2}{6}\Big) = s\Big(-\frac{r}{2} - \frac{s}{4} + \frac{s+r}{6}\Big) = \frac{s}{12}\big(-4r - s\big).$$

So $V = 0$ iff $s = -4r$. Substituting into $s^2 = s + r$: $16r^2 = -4r + r = -3r$, so $r = 0$ or $r = -\frac{3}{16}$. The nontrivial solution is

$$\boxed{r_c = -\tfrac{3}{16}}, \qquad s = \tfrac{3}{4}, \quad x^* = \pm\tfrac{\sqrt{3}}{2}.$$

Check: it lies in $(-\frac{1}{4}, 0)$, where five fixed points exist, and $s_+ = \frac{1}{2}\big(1 + \sqrt{1 - \frac{3}{4}}\big) = \frac{3}{4}$. ✓

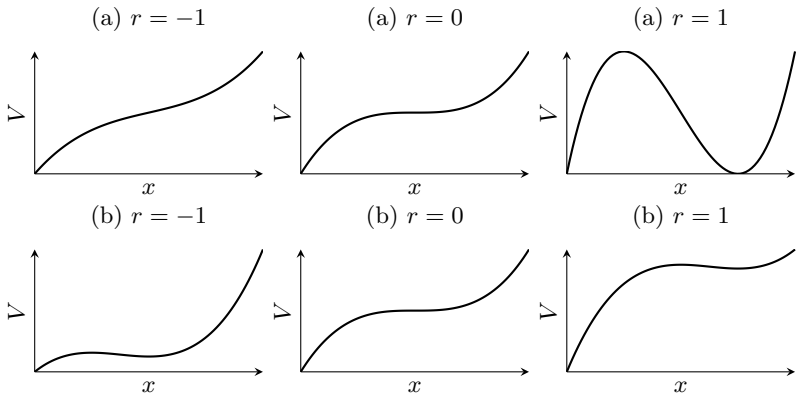
At $r = r_c$ the system is equally likely to be found in any of the three minima, like water and ice at the freezing point.

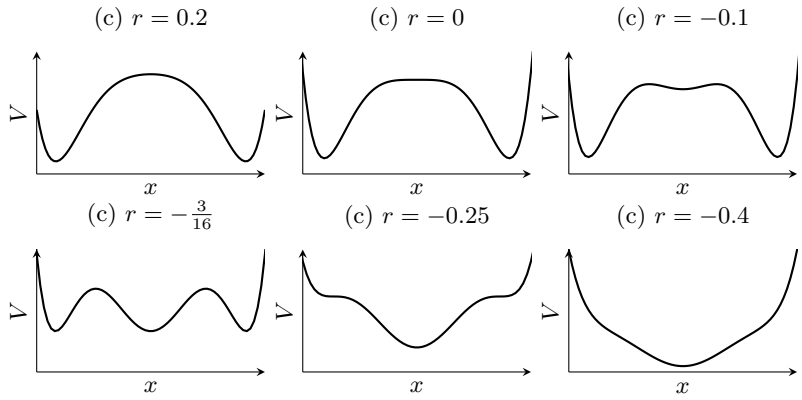
Exercise 3.4.16 (Potentials) Sketch $V(x)$ (with $\dot{x} = -V'$) as a function of r for (a) $\dot{x} = r - x^2$, (b) $\dot{x} = rx - x^2$, (c) $\dot{x} = rx + x^3 - x^5$.

Solution. **(a)** $V = -rx + \frac{1}{3}x^3$, $V' = x^2 - r$. For $r < 0$, $V' > 0$: V is strictly increasing, with no extrema. At $r = 0$, $V = \frac{1}{3}x^3$ has a single inflection point at 0. For $r > 0$: a local maximum at $x = -\sqrt{r}$ (unstable) and a local minimum at $x = +\sqrt{r}$ (stable).

(b) $V = -\frac{r}{2}x^2 + \frac{1}{3}x^3$, $V' = x(x - r)$. For $r < 0$: minimum at 0, maximum at $x = r$. At $r = 0$: inflection at 0. For $r > 0$: maximum at 0, minimum at $x = r$. The extrema exchange roles as r passes 0.

(c) $V = -\frac{r}{2}x^2 - \frac{1}{4}x^4 + \frac{1}{6}x^6$. For $r > 0$: a double well (maximum at 0, minima at $\pm\sqrt{s_+}$). At $r = 0$ the maximum at 0 is very flat ($V \approx -x^4/4$). For $-\frac{1}{4} < r < 0$: three wells (minimum at 0, minima at $\pm\sqrt{s_+}$, separated by maxima at $\pm\sqrt{s_-}$); the outer wells are deeper than the central one for $r > -\frac{3}{16}$, equal at $r = -\frac{3}{16}$, shallower for $r < -\frac{3}{16}$. At $r = -\frac{1}{4}$ the outer wells disappear into inflection points; for $r < -\frac{1}{4}$ only the central well remains.





3.5 Overdamped Bead on a Rotating Hoop

Setup used below (from the text). With $\gamma = r\omega^2/g$, $T = b/(mg)$, $\tau = t/T$ and $\varepsilon = m^2gr/b^2$, the bead equation becomes

$$\varepsilon \frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} + \frac{\mathrm{d}\phi}{\mathrm{d}\tau} = f(\phi), \qquad f(\phi) = \sin\phi\,(\gamma\cos\phi - 1),$$

and the overdamped limit is $\dfrac{\mathrm{d}\phi}{\mathrm{d}\tau} = f(\phi)$ (Eq. 3.5.7).

Exercise 3.5.1 Why can't the bead have an equilibrium with $\phi > \pi/2$ (other than the top, $\phi = \pi$)?

Solution. For $\pi/2 < \phi < \pi$ the bead is on the upper half of the hoop. Gravity has a tangential component pulling it *down* the hoop, toward smaller ϕ. The centrifugal force is horizontal and points away from the axis; its tangential component is $mr\omega^2 \sin\phi\cos\phi < 0$ there, because $\cos\phi < 0$. This also points toward smaller ϕ (moving to smaller ϕ takes the bead farther from the axis, since $\rho = r\sin\phi$ decreases with ϕ on this half). Both tangential forces point the same way, so they cannot cancel. (At $\phi = \pi$ both vanish because $\sin\phi = 0$, which is why the top is still a fixed point.)

Exercise 3.5.2 Linear stability analysis of all fixed points of Eq. (3.5.7).

Solution. $f(\phi) = \sin\phi(\gamma\cos\phi - 1)$, so

$$f'(\phi) = \cos\phi(\gamma\cos\phi - 1) - \gamma\sin^2\phi.$$

- $\phi^* = 0$: $f'(0) = \gamma - 1$. Stable for $\gamma < 1$, unstable for $\gamma > 1$.
- $\phi^* = \pi$: $f'(\pi) = (-1)(-\gamma - 1) = \gamma + 1 > 0$. Always unstable.
- $\phi^* = \pm\cos^{-1}(1/\gamma)$ (exist for $\gamma > 1$): here $\gamma\cos\phi^* = 1$, so $f' = -\gamma\sin^2\phi^* = -\gamma\big(1 - \frac{1}{\gamma^2}\big) = -\dfrac{\gamma^2 - 1}{\gamma} < 0$. Stable.

Thus the bottom is stable for $\gamma < 1$ and loses stability at $\gamma = 1$ in a supercritical pitchfork, giving birth to the two stable symmetric equilibria; the top is unstable for all γ. This agrees with Figure 3.5.6.

Exercise 3.5.3 Show that Eq. (3.5.7) reduces to $\dfrac{\mathrm{d}\phi}{\mathrm{d}\tau} = A\phi - B\phi^3 + O(\phi^5)$ near $\phi = 0$; find A and B.

Solution. Using $\sin\phi = \phi - \frac{\phi^3}{6} + \dots$ and $\cos\phi = 1 - \frac{\phi^2}{2} + \dots$,

$$\gamma\cos\phi - 1 = (\gamma - 1) - \tfrac{\gamma}{2}\phi^2 + O(\phi^4),$$

so

$$f(\phi) = \Big(\phi - \tfrac{\phi^3}{6}\Big)\Big((\gamma-1) - \tfrac{\gamma}{2}\phi^2\Big) + O(\phi^5) = (\gamma-1)\phi - \Big[\tfrac{\gamma}{2} + \tfrac{\gamma-1}{6}\Big]\phi^3 + O(\phi^5).$$

Hence

$$\boxed{A = \gamma - 1, \qquad B = \frac{4\gamma - 1}{6}.}$$

At $\gamma = 1$, $B = \frac{1}{2} > 0$: a supercritical pitchfork.

Exercise 3.5.4 (Bead on a horizontal wire) Mass m on a horizontal wire, attached to a spring (relaxed length L_0, constant k) whose support point is a distance h from the wire; viscous damping $b\dot{x}$.

Solution. **(a)** Let x be the position along the wire measured from the point closest to the support. The spring length is $L = \sqrt{x^2 + h^2}$, the spring force has magnitude $k(L - L_0)$ and points toward the support, so its component along the wire is $-k(L - L_0)\,x/L$. Newton's law:

$$m\ddot{x} = -b\dot{x} - kx\Big(1 - \frac{L_0}{\sqrt{x^2 + h^2}}\Big).$$

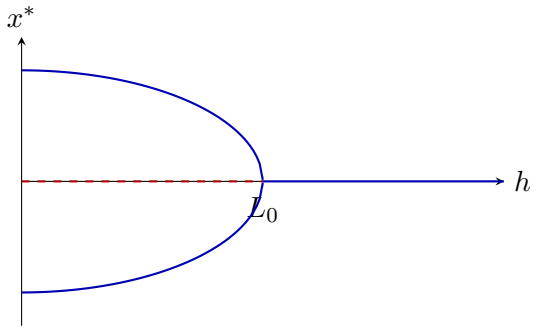
(b) Equilibria ($\dot{x} = \ddot{x} = 0$): $x^* = 0$ always, and $\sqrt{x^2 + h^2} = L_0$, i.e.

$$x^* = \pm\sqrt{L_0^2 - h^2} \quad \text{if } L_0 > h.$$

(c) For $m = 0$: $b\dot{x} = f(x) := -kx\Big(1 - \dfrac{L_0}{\sqrt{x^2 + h^2}}\Big)$.

- $f'(0) = -k\big(1 - L_0/h\big)$: $x^* = 0$ is stable if $L_0 < h$ (spring stretched) and unstable if $L_0 > h$ (spring compressed).
- At $x^* = \pm\sqrt{L_0^2 - h^2}$: differentiating, $f'(x) = -k\Big[1 - \dfrac{L_0}{L} + \dfrac{L_0 x^2}{L^3}\Big]$, and at $L = L_0$ this is $-kx^2/L_0^2 < 0$. Stable.

So there is a supercritical pitchfork at $h = L_0$. (Plotting x^* versus h, the nonzero branches trace out the semicircle $x^2 + h^2 = L_0^2$.)



(d) Compare $m\ddot{x}$ with $b\dot{x}$ using the slow time scale $T = b/k$ (from $b\dot{x} \sim kx$). Then $m\ddot{x} \sim mx/T^2$ and $b\dot{x} \sim kx$, so the ratio is

$$\frac{m/T^2}{k} = \frac{mk}{b^2}.$$

The inertial term is negligible when $mk/b^2 \ll 1$, i.e. $m \ll b^2/k$ (heavy damping or small mass). The sense in which it is negligible: the velocity relaxes to its terminal value $f(x)/b$ on the fast

time scale m/b, which is $\ll$ the slow time scale b/k; after a brief initial transient the motion follows $b\dot{x} = f(x)$.

Exercise 3.5.5 (Time scale of the rapid transient) $\varepsilon\,\phi'' + \phi' = f(\phi)$.

Solution. **(a)** During the rapid transient ϕ' is large and f is negligible by comparison, so $\varepsilon\phi'' + \phi' \approx 0$, giving $\phi' \propto e^{-\tau/\varepsilon}$. Hence $\tau_{\text{fast}} \sim \varepsilon$. Converting back with $t = T\tau$, $T = b/(mg)$:

$$T_{\text{fast}} \sim \varepsilon\,\frac{b}{mg} = \frac{m^2 g r}{b^2}\cdot\frac{b}{mg} = \frac{mr}{b}.$$

This is also read directly off $mr\ddot{\phi} = -b\dot{\phi}$ (rate $b/(mr)$). It does not involve g or ω.

(b) Rescale time by $T = T_{\text{fast}} = mr/b$ in Eq. (3): the coefficients become $\dfrac{r}{gT^2} = \dfrac{b^2}{m^2 g r} = \dfrac{1}{\varepsilon}$ and $\dfrac{b}{mgT} = \dfrac{1}{\varepsilon}$. With $s = t/T_{\text{fast}}$:

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}s^2} + \frac{\mathrm{d}\phi}{\mathrm{d}s} = \varepsilon f(\phi).$$

So on this time scale the gravity and centrifugal terms (all of f) are negligible, being $O(\varepsilon)$; only inertia and damping matter.

(c)

$$\frac{T_{\text{fast}}}{T_{\text{slow}}} = \frac{mr/b}{b/(mg)} = \frac{m^2 g r}{b^2} = \varepsilon \ll 1.$$

Exercise 3.5.6 (A model problem about singular limits) $\varepsilon\ddot{x} + \dot{x} + x = 0$, $x(0) = 1$, $\dot{x}(0) = 0$.

Solution. **(a)** The characteristic equation $\varepsilon\lambda^2 + \lambda + 1 = 0$ has roots $\lambda_{1,2} = \dfrac{-1 \pm \sqrt{1-4\varepsilon}}{2\varepsilon}$.

- $0 < \varepsilon < \frac{1}{4}$ (real roots):

  $$x(t) = \frac{\lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t}}{\lambda_2 - \lambda_1}$$

  (check: $x(0) = 1$, $\dot{x}(0) = (\lambda_2\lambda_1 - \lambda_1\lambda_2)/(\lambda_2 - \lambda_1) = 0$), with $\lambda_1 = \frac{-1+\sqrt{1-4\varepsilon}}{2\varepsilon}$, $\lambda_2 = \frac{-1-\sqrt{1-4\varepsilon}}{2\varepsilon}$.
- $\varepsilon = \frac{1}{4}$: $x(t) = (1+2t)e^{-2t}$.
- $\varepsilon > \frac{1}{4}$: $x(t) = e^{-t/(2\varepsilon)}\Big[\cos\omega t + \dfrac{1}{2\varepsilon\omega}\sin\omega t\Big]$, $\omega = \dfrac{\sqrt{4\varepsilon - 1}}{2\varepsilon}$.

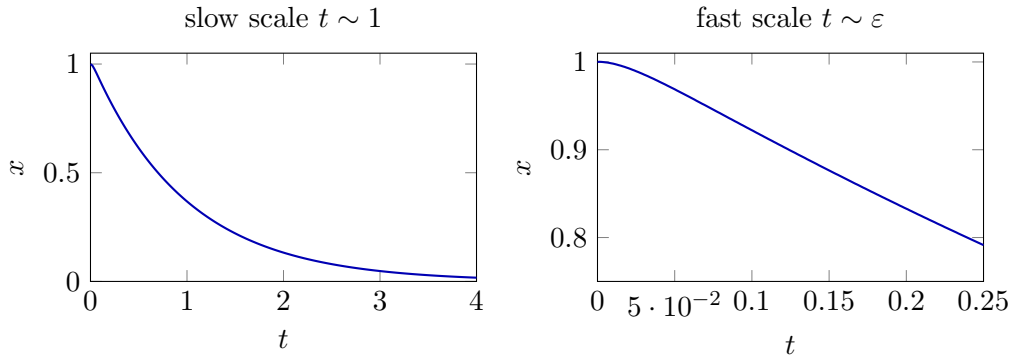
(b) For $\varepsilon \ll 1$, $\sqrt{1-4\varepsilon} \approx 1 - 2\varepsilon - 2\varepsilon^2$, so

$$\lambda_1 \approx -1 - \varepsilon, \qquad \lambda_2 \approx -\frac{1}{\varepsilon} + 1.$$

The two time scales are widely separated: a slow one $t \sim 1$ and a fast one $t \sim \varepsilon$. Indeed,

$$x(t) \approx e^{-t} - \varepsilon\, e^{-t/\varepsilon} \qquad (\varepsilon \ll 1).$$

(c) The graph (shown for $\varepsilon = 0.02$) is essentially the slow exponential e^{-t}. The fast term is an $O(\varepsilon)$-amplitude transient on the scale $t \sim \varepsilon$, during which the slope goes from $\dot{x}(0) = 0$ to ≈ -1 (see the magnified plot on the right).



(d) Setting $\varepsilon = 0$ gives $\dot{x} + x = 0$, whose solutions $x = x_0 e^{-t}$ can satisfy only *one* of the two initial conditions. The singular limit therefore loses the initial layer: it is a good approximation (error $O(\varepsilon)$ in x) once $t \gg \varepsilon$, but it is wrong for $t \lesssim \varepsilon$ (e.g. its slope at $t = 0$ is -1 instead of 0), and it cannot impose the initial condition $\dot{x}(0) = 0$.

(e) *Mechanical:* $m\ddot{x} + b\dot{x} + kx = 0$. With $\tau = kt/b$ this becomes $\varepsilon x'' + x' + x = 0$ with $\varepsilon = mk/b^2$; $\varepsilon \ll 1$ means heavy damping or small mass (strongly overdamped), so inertia matters only briefly. *Electrical:* the LRC circuit $L\ddot{Q} + R\dot{Q} + Q/C = 0$ with $\tau = t/(RC)$ gives $\varepsilon = L/(R^2C)$; $\varepsilon \ll 1$ means the inductive time L/R is much shorter than the RC relaxation time, so inductance is negligible after a short transient. In both cases ε is the ratio of the fast (inertial) time scale to the slow (relaxation) one.

Exercise 3.5.7 (Nondimensionalizing the logistic equation) $\dot{N} = rN(1 - N/K)$, $N(0) = N_0$.

Solution. **(a)** $[r] = \text{time}^{-1}$, $[K] = [N_0] =$ the units of N (population or density).

(b) Let $x = N/K$, $\tau = rt$, $x_0 = N_0/K$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \frac{1}{rK}\dot{N} = x(1 - x), \qquad x(0) = x_0.$$

(c) Let $u = N/N_0$ and again $\tau = rt$. Then $\dfrac{\mathrm{d}u}{\mathrm{d}\tau} = u(1 - x_0 u)$, with $u(0) = 1$ for every choice of parameters; the single dimensionless group $x_0 = N_0/K$ now sits in the equation.

(d) In (b) the equation is universal (no parameters), so all solutions are members of one family and the parameter dependence is entirely in the initial condition x_0; this is convenient for studying the flow and the fixed points. In (c) the initial condition is fixed, which makes the early exponential growth $u \approx e^{\tau}$ (rate r) and the dependence of the saturation level $1/x_0$ on N_0/K explicit, and is convenient for comparing curves started from the same initial value.

Exercise 3.5.8 (Nondimensionalizing the subcritical pitchfork) $\dot{u} = au + bu^3 - cu^5$, $b, c > 0$. Show it can be written $\dfrac{\mathrm{d}x}{\mathrm{d}\tau} = rx + x^3 - x^5$ with $x = u/U$, $\tau = t/T$.

Solution. Substitute $u = Ux$, $t = T\tau$:

$$\frac{U}{T}\frac{\mathrm{d}x}{\mathrm{d}\tau} = aUx + bU^3x^3 - cU^5x^5 \implies \frac{\mathrm{d}x}{\mathrm{d}\tau} = aT\,x + bU^2T\,x^3 - cU^4T\,x^5.$$

Require $bU^2T = 1$ and $cU^4T = 1$. Dividing, $cU^2/b = 1$, so

$$\boxed{U = \sqrt{\frac{b}{c}}, \qquad T = \frac{1}{bU^2} = \frac{c}{b^2}, \qquad r = aT = \frac{ac}{b^2}.}$$

3.6 Imperfect Bifurcations and Catastrophes

Exercise 3.6.1 Does Figure 3.6.3b correspond to $h > 0$ or $h < 0$?

Solution. For $\dot{x} = h + rx - x^3$, a fixed point with $r < 0$ and small $|x|$ satisfies $x^* \approx h/(-r)$, which has the sign of h. In Figure 3.6.3b the continuous branch (which passes smoothly through $r = 0$) lies at $x > 0$ for $r < 0$, so the figure corresponds to $\boldsymbol{h > 0}$. (For $h < 0$ the picture is reflected: $(x, h) \to (-x, -h)$ is a symmetry of the equation.)

Exercise 3.6.2 (Imperfect transcritical bifurcation) $\dot{x} = h + rx - x^2$.

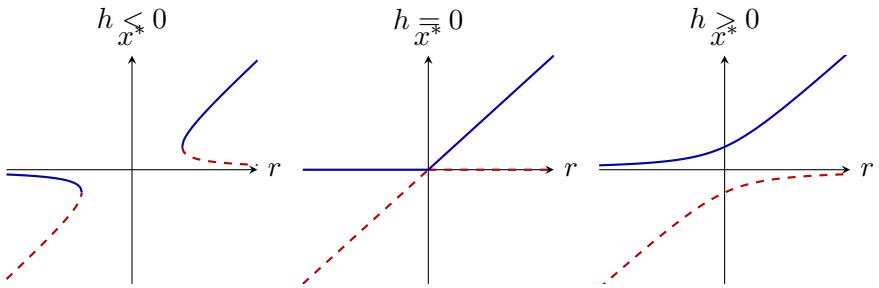
Solution. Fixed points: $x^2 - rx - h = 0$, so

$$x_\pm = \frac{r \pm \sqrt{r^2 + 4h}}{2}, \qquad f'(x) = r - 2x, \quad f'(x_\pm) = \mp\sqrt{r^2 + 4h}.$$

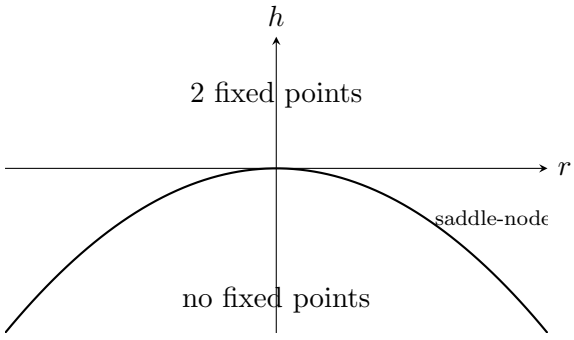
Thus (when real) x_+ is stable and x_- is unstable.

(a) Bifurcation diagrams:

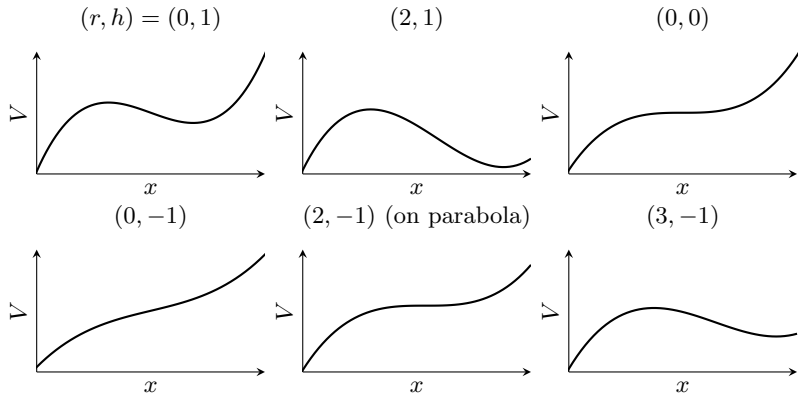
- $h = 0$: the usual transcritical bifurcation, $x^* = 0$ and $x^* = r$ exchanging stability at $r = 0$.
- $h > 0$: $r^2 + 4h > 0$ for all r, so two fixed points exist for every r. Since $x_+ x_- = -h < 0$ they have opposite signs, so $x_+ > 0 > x_-$ and the two branches never meet: the transcritical crossing is "unfolded" into two disjoint smooth curves.
- $h < 0$: fixed points exist only for $|r| \geq 2\sqrt{-h}$; for $|r| < 2\sqrt{-h}$ there are none ($\dot{x} < 0$ everywhere). At $r = \pm 2\sqrt{-h}$ there are saddle-node bifurcations: the transcritical splits into two saddle-nodes.



(b) The number of fixed points changes where $r^2 + 4h = 0$, i.e. on the parabola $h = -r^2/4$ in the (r, h) plane. Above it ($h > -r^2/4$) there are two fixed points (one stable, one unstable); below it there are none. On the parabola there is a saddle-node bifurcation, except at the vertex $(r, h) = (0, 0)$, which is the (codimension-2) transcritical point.



(c) The potential is $V(x) = -hx - \frac{r}{2}x^2 + \frac{1}{3}x^3$ (so that $\dot{x} = -V'$). In the region with two fixed points, V is a cubic with a local minimum at x_+ (stable) and a local maximum at x_- (unstable). On the parabola the extremum pair merges into an inflection point; below it ($r^2 + 4h < 0$) $V' > 0$ and V is monotonic with no equilibria.



Exercise 3.6.3 (A perturbation to the supercritical pitchfork) $\dot{x} = rx + ax^2 - x^3$.

Solution. **(a)** $\dot{x} = x(r + ax - x^2)$. Fixed points: $x^* = 0$ and

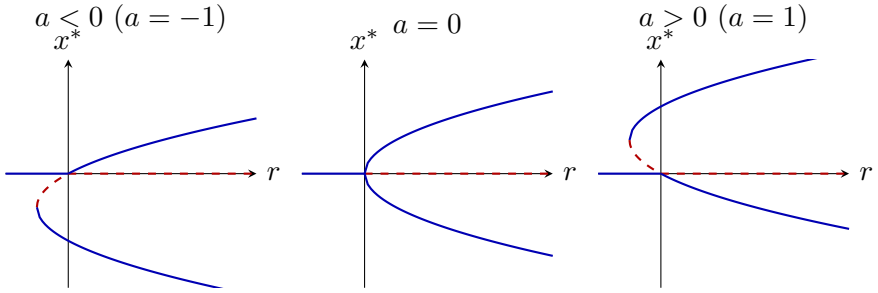
$$x_\pm = \frac{a \pm \sqrt{a^2 + 4r}}{2} \quad \text{(real iff } r \geq -a^2/4\text{)}.$$

Stability: $f' = r + 2ax - 3x^2$. At 0: $f' = r$. At $x_\pm$ (use $x^2 = ax + r$): $f' = -2r - ax$; writing $s = \sqrt{a^2 + 4r}$, this is $-\frac{s}{2}(s \pm a)$.

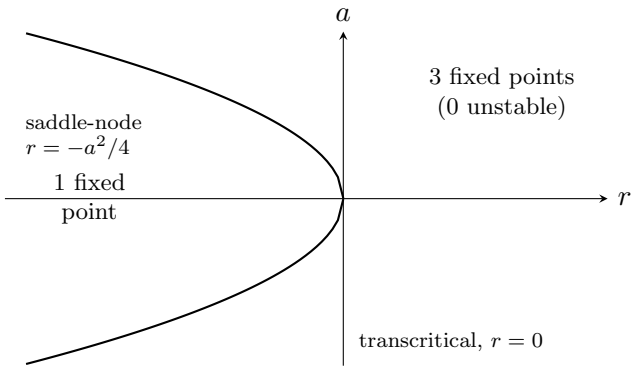
At $r = 0$ the root $x_\mp$ (the one with sign opposite to a) passes through 0: since $x_\pm = \frac{1}{2}(a \pm |a|)$ at $r = 0$. So $x = 0$ and that branch cross transcritically at $r = 0$ for $a \neq 0$, while the pair $x_\pm$ is born at the saddle-node $r = -a^2/4$, $x = a/2$. For $a > 0$:

- $r < -a^2/4$: only $x = 0$, stable.
- $-a^2/4 < r < 0$: 0 stable, $x_- \in (0, a/2)$ unstable, x_+ stable.
- $r > 0$: $x_- < 0$ stable, 0 unstable, x_+ stable.

The case $a < 0$ is the mirror image under $x \to -x$ (the equation is invariant under $(x, a) \to (-x, -a)$). So the qualitatively different bifurcation diagrams are: the pitchfork ($a = 0$), and the "saddle-node plus transcritical" diagram, which comes in two mirror-image versions for $a > 0$ and $a < 0$.



(b) In the (r, a) plane the boundaries are the parabola $r = -a^2/4$ (saddle-node bifurcations) and the line $r = 0$ (transcritical, for $a \neq 0$). Left of the parabola there is one fixed point; to its right there are three. Between the parabola and the line $r = 0$ the three fixed points are ordered (stable 0, unstable x_-, stable x_+); to the right of $r = 0$ they are reordered as (x_- stable, 0 unstable, x_+ stable): in both regions the pattern along the x-axis is S–U–S, and crossing $r = 0$ merely makes two fixed points pass through each other and swap stability. The parabola and the line touch at the origin, where (for $a = 0$) the pitchfork occurs.



(Between the parabola and the a-axis, i.e. $-a^2/4 < r < 0$, there are also three fixed points, with 0 stable.)

Exercise 3.6.4 (Imperfect saddle-node) What happens if you add a small imperfection to a system with a saddle-node bifurcation?

Solution. Essentially nothing qualitative: the saddle-node is *structurally stable* (generic). For the normal form $\dot{x} = r + x^2$, adding a constant imperfection h gives $\dot{x} = (r + h) + x^2$, which is the same system with the bifurcation point shifted from $r = 0$ to $r = -h$. More generally, for $\dot{x} = f(x, r) + \varepsilon g(x, r)$ the saddle-node conditions $f = 0$, $f_x = 0$ are two equations in the two unknowns (x_c, r_c) whose Jacobian $\begin{pmatrix} f_x & f_r \\ f_{xx} & f_{xr} \end{pmatrix} = \begin{pmatrix} 0 & f_r \\ f_{xx} & f_{xr} \end{pmatrix}$ has determinant $-f_r f_{xx} \neq 0$ at a saddle-node. By the implicit function theorem the solution (x_c, r_c) persists and moves smoothly with ε. In contrast, the transcritical and pitchfork bifurcations rely on special structure (a fixed point forced at $x = 0$, or a reflection symmetry), so they are destroyed (unfolded) by generic perturbations.

Exercise 3.6.5 (Bead on a tilted wire) Equilibria, dimensionless form, catastrophe.

Solution. **(a)** Let x be measured along the wire (downhill positive) from the point closest to the spring's support, which is at distance a from the wire. Gravity contributes $mg\sin\theta$ along the wire. The spring (length $\sqrt{x^2 + a^2}$) contributes $-k\big(\sqrt{x^2 + a^2} - L_0\big)\, x/\sqrt{x^2 + a^2}$. At equilibrium these balance:

$$mg\sin\theta = kx\Big(1 - \frac{L_0}{\sqrt{x^2 + a^2}}\Big).$$

(b) Put $x = au$ and divide by ka:

$$\frac{mg\sin\theta}{ka} = u\Big(1 - \frac{L_0/a}{\sqrt{1 + u^2}}\Big).$$

With $R = \dfrac{L_0}{a}$, $h = \dfrac{mg\sin\theta}{ka}$, $u = \dfrac{x}{a}$, and dividing by u,

$$\boxed{1 - \frac{h}{u} = \frac{R}{\sqrt{1 + u^2}}.}$$

(c) Plot the left side $1 - h/u$ and the right side $R/\sqrt{1 + u^2}$ against u (take $h > 0$). The right side is an even bump of height R decaying to 0. The left side is $1 - h/u$: for $u > 0$ it increases from $-\infty$ to 1; for $u < 0$ it equals $1 + h/|u|$, decreasing from $+\infty$ to 1.

- $R < 1$: the bump never exceeds 1. For $u < 0$ the left side is $> 1 >$ right side: no intersection. For $u > 0$ an increasing curve meets a decreasing one exactly once. *One equilibrium* (downhill), for every h.

- $R > 1$: for $u > 0$ there is again exactly one intersection (downhill equilibrium). For $u < 0$ the left side starts at $+\infty$ and decays to 1, while the bump rises to $R > 1$ near $u = 0$: for small h they intersect twice, for large h not at all. So there are *one or three* equilibria: three when $|h|$ is small enough (the extra pair is on the uphill side), one when $|h|$ is large.

(d) Expand the equilibrium equation $h = u - \dfrac{Ru}{\sqrt{1+u^2}}$ for small u using $(1+u^2)^{-1/2} \approx 1 - \frac{1}{2}u^2$:

$$h \approx u - Ru + \tfrac{R}{2}u^3 = -(R-1)u + \tfrac{R}{2}u^3.$$

With $r = R - 1$ small, $R \approx 1$, this is

$$h + ru - \tfrac{1}{2}u^3 \approx 0.$$

(e) Saddle-node bifurcations of $g(u) = h + ru - \frac{1}{2}u^3 = 0$: tangency requires $g'(u) = r - \frac{3}{2}u^2 = 0$, i.e. $u = \pm\sqrt{2r/3}$ (so $r > 0$). Then $h = -ru + \frac{1}{2}u^3 = -\frac{2}{3}ru$, giving

$$h = \mp\Big(\frac{2r}{3}\Big)^{3/2}, \quad \text{i.e.} \quad 27h^2 = 8r^3.$$

This is a pair of curves meeting tangentially at the cusp $(r, h) = (0, 0)$.

(f) Exactly: write $F(u, R) = u - \dfrac{Ru}{\sqrt{1+u^2}}$. A saddle-node requires $F = h$ and $\partial F/\partial u = 0$. Since $\dfrac{\partial}{\partial u}\dfrac{u}{\sqrt{1+u^2}} = (1+u^2)^{-3/2}$,

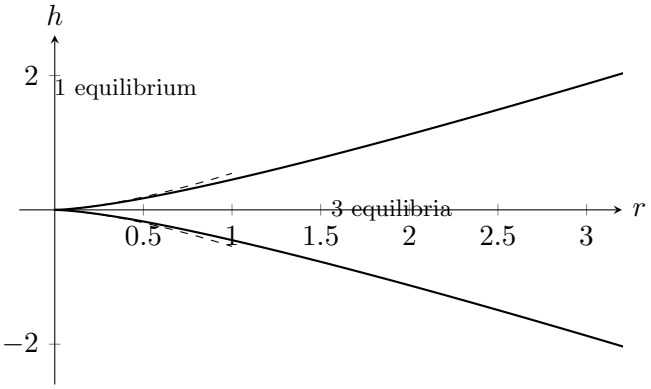
$$\partial_u F = 1 - R(1+u^2)^{-3/2} = 0 \;\Rightarrow\; R = (1+u^2)^{3/2}.$$

Then $h = u - (1+u^2)^{3/2}\dfrac{u}{(1+u^2)^{1/2}} = u - u(1+u^2) = -u^3$. So

$$\boxed{h(u) = -u^3, \qquad R(u) = (1+u^2)^{3/2}, \qquad -\infty < u < \infty.}$$

Check against (e): for small u, $r = R - 1 \approx \frac{3}{2}u^2$, so $u = \pm\sqrt{2r/3}$ and $h = -u^3 = \mp(2r/3)^{3/2}$. ✓

(g) The exact bifurcation curves in the (r, h) plane, $r = R - 1 = (1+u^2)^{3/2} - 1$, $h = -u^3$ (solid), together with the small-(r, h) approximation from (e) (dashed). Inside the cusp there are three equilibria; outside, one.



(h) In dimensional terms: $R = L_0/a$ compares the relaxed spring length with the support distance, and $h = mg\sin\theta/(ka)$ measures the tilt (gravity along the wire relative to the spring stiffness). If $L_0 < a$ ($R < 1$) the spring is always stretched and there is a single, smoothly varying equilibrium for any tilt: no catastrophe. If $L_0 > a$ ($R > 1$), the untilted wire has three

equilibria (unstable centre, two stable symmetric ones: a supercritical pitchfork at $L_0 = a$). For a small tilt there are still three equilibria: the downhill one plus a stable/unstable pair on the uphill side. As the tilt grows, the uphill stable equilibrium collides with the unstable one and vanishes in a saddle-node at

$$\sin\theta_c = \frac{ka}{mg}\left[\left(\frac{L_0}{a}\right)^{2/3} - 1\right]^{3/2}$$

(from $|h| = |u|^3$, $(1+u^2) = R^{2/3}$). A bead resting at the uphill equilibrium then jumps catastrophically to the downhill one. Reducing the tilt again does not bring it back (hysteresis): the bead stays downhill.

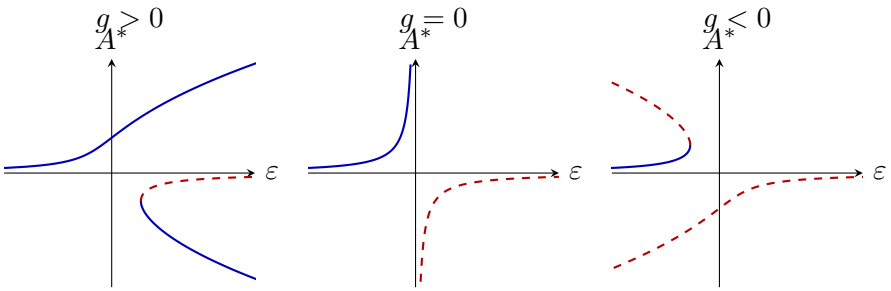
Exercise 3.6.6 (Patterns in fluids) Amplitude equation $\tau\dot{A} = \varepsilon A - gA^3$ (supercritical, $g > 0$) or $\tau\dot{A} = \varepsilon A - gA^3 - kA^5$ (subcritical, $g < 0$, $k > 0$).

Solution. **(a)** Steady state: $A^{*2} = \varepsilon/g$, so $A^* = \sqrt{\varepsilon/g} \propto \varepsilon^{1/2}$: the Landau equation predicts $\beta = 0.5$, in agreement with the measured 0.50 ± 0.01.

(b) For $g = 0$: $\tau\dot{A} = \varepsilon A - kA^5$, so $A^{*4} = \varepsilon/k$ and $A^* = (\varepsilon/k)^{1/4} \propto \varepsilon^{1/4}$: $\beta = \frac{1}{4}$ at a tricritical point.

(c) Now $\tau\dot{A} = h + \varepsilon A - gA^3$, $h > 0$ small. The steady states satisfy $\varepsilon = gA^2 - h/A$ (for $A \neq 0$), and the stability is that of $f = h + \varepsilon A - gA^3$, with $f' = \varepsilon - 3gA^2 = -2gA^2 - h/A$ at a fixed point. (The diagrams below use $h = 0.1$ and $|g| = 1$.)

- $g > 0$ (imperfect supercritical pitchfork): one continuous, stable branch with $A^* > 0$ passing smoothly through $\varepsilon = 0$ at $A^* \approx (h/g)^{1/3}$, plus a disconnected piece for $A^* < 0$, which consists of an unstable part (small $|A|$) and a stable part, joined at a saddle-node.
- $g = 0$: $A^* = -h/\varepsilon$ (a hyperbola). The branch $\varepsilon < 0$, $A^* > 0$ is stable ($f' = \varepsilon < 0$) and diverges as $\varepsilon \to 0^-$; the branch $\varepsilon > 0$, $A^* < 0$ is unstable. (A quintic term is needed to saturate the amplitude.)
- $g < 0$ (imperfect subcritical pitchfork), writing $G = |g|$: $\varepsilon = -h/A - GA^2$. The stable branch has $0 < A < A_0 = (h/2G)^{1/3}$ and exists for $\varepsilon < \varepsilon_s = -\frac{3}{2}(2G)^{1/3}h^{2/3} < 0$, where it terminates in a saddle-node with the unstable branch $A > A_0$. A third, unstable branch lies at $A < 0$.



Comparing with the data of Aitta et al. (Fig. 2 of their paper, or Fig. 15 of Ahlers 1989) I leave to the reader, since it requires the published figures.

(d) Consider the imperfect subcritical case with the quintic term, $\tau\dot{A} = h + \varepsilon A + GA^3 - kA^5$ with ε switched at $t = 0$ from a negative value to $\varepsilon_f > 0$ (so A starts near 0). For A small, $\dot{A}$ is positive and, if ε_f is small, nearly flat: $\tau\dot{A} \approx h + \varepsilon_f A$ is small over a wide range of A. Plotting $\dot{A}$ versus A, the curve stays close to the axis (a near-bottleneck, the ghost of the saddle-node) so A creeps slowly upward for a long time, giving the plateau. Only when A has grown enough for the cubic term GA^3 to dominate does $\dot{A}$ become large, after which A shoots up rapidly to the large-amplitude fixed point set by the quintic term. For large ε_f, $\dot{A}$ is large and positive for all A (no bottleneck), so A rises almost directly to its steady state.

Exercise 3.6.7 (Simple model of a magnet) Mean-field equation $h = T\tanh^{-1} m - Jnm$.

Solution. **(a)** Treat $m \in (-1, 1)$ as signed. Define $H(m) = T\tanh^{-1} m - Jnm$; solutions of the equilibrium condition for a given h are the intersections of the graph of $H(m)$ with the horizontal line at height h. The function H is odd and $H \to \pm\infty$ as $m \to \pm 1$. Its derivative is

$$H'(m) = \frac{T}{1-m^2} - Jn.$$

- $T > Jn$: $H' > 0$ everywhere, so H is monotonic; for every h there is exactly one solution (a paramagnet: $m = 0$ when $h = 0$).
- $T < Jn$: $H'(0) = T - Jn < 0$; H is S-shaped with local extrema at $m = \pm m_e$, $m_e^2 = 1 - T/(Jn)$. For $|h| < h_c(T) = |H(m_e)|$ there are three solutions (two stable, one unstable), for $|h| > h_c$ just one. At $h = 0$ the solutions are $m = 0$ and $m = \pm m^*$: two symmetry-related ferromagnetic states.

In the language of Section 3.6, h is the imperfection parameter and T plays the role of $-r$: the model has an imperfect pitchfork with hysteresis (a first-order jump of m when h is swept through $\pm h_c$ at fixed $T < Jn$).

(b) For $h = 0$: $\tanh^{-1} m = (Jn/T)m$. Expanding, $m + \frac{1}{3}m^3 + \cdots = (Jn/T)m$. A nonzero solution exists iff $Jn/T > 1$. So

$$\boxed{T_c = Jn}.$$

Near T_c: $m^{*2} \approx 3\big(\frac{Jn}{T} - 1\big) = 3\,\frac{T_c - T}{T}$, so $m^* \propto (T_c - T)^{1/2}$ (a supercritical pitchfork in m as T decreases through T_c).

3.7 Insect Outbreak

Setup used below (from the text). The dimensionless budworm model is

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = rx\Big(1 - \frac{x}{k}\Big) - \frac{x^2}{1+x^2} \qquad (r, k > 0),$$

and its saddle-node bifurcation curves are given parametrically (for $x > 1$) by

$$r(x) = \frac{2x^3}{(1+x^2)^2}, \qquad k(x) = \frac{2x^3}{x^2 - 1}.$$

Exercise 3.7.1 Show that the fixed point $x^* = 0$ is always unstable for Eq. (3.7.3).

Solution. With $f(x) = rx(1 - x/k) - \dfrac{x^2}{1+x^2}$,

$$f'(x) = r\Big(1 - \frac{2x}{k}\Big) - \frac{2x}{(1+x^2)^2}, \qquad f'(0) = r > 0.$$

Since the dimensionless growth rate is $r > 0$, $x^* = 0$ is always unstable. (Intuitively, predation is $O(x^2)$ and negligible for small x, so the population grows exponentially.)

Exercise 3.7.2 (Bifurcation curves for the insect outbreak model) (a) Limits of $r(x)$, $k(x)$ as $x \to 1^+$ and $x \to \infty$. (b) Exact r, k, x at the cusp point.

Solution. **(a)**

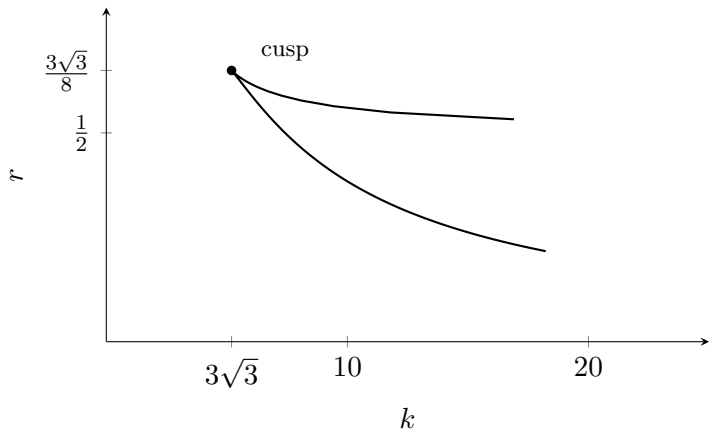
- $x \to 1^+$: $r \to \dfrac{2}{4} = \dfrac{1}{2}$, and $k = \dfrac{2x^3}{(x-1)(x+1)} \to +\infty$ (like $1/(x-1)$).
- $x \to \infty$: $r \approx \dfrac{2x^3}{x^4} = \dfrac{2}{x} \to 0$, and $k \approx 2x \to \infty$.

(b) A cusp occurs where both $r'(x)$ and $k'(x)$ vanish. Differentiating,

$$r'(x) = \frac{2x^2(3-x^2)}{(1+x^2)^3}, \qquad k'(x) = \frac{2x^2(x^2-3)}{(x^2-1)^2}.$$

Both vanish at $x = \sqrt{3}$ (for $x > 1$). Then

$$r_c = \frac{2 \cdot 3\sqrt{3}}{16} = \frac{3\sqrt{3}}{8} \approx 0.65, \qquad k_c = \frac{2 \cdot 3\sqrt{3}}{2} = 3\sqrt{3} \approx 5.20, \qquad x_c = \sqrt{3}.$$



Exercise 3.7.3 (A model of a fishery) $\dot{N} = rN(1 - N/K) - H$.

Solution. **(a)** Let $x = N/K$, $\tau = rt$, $h = H/(rK)$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \frac{1}{rK}\dot{N} = x(1-x) - h.$$

(b) The vector field $f(x) = x - x^2 - h$ is a downward parabola with maximum $\frac{1}{4} - h$ at $x = \frac{1}{2}$:

- $h < \frac{1}{4}$: two fixed points $x_\pm = \frac{1}{2}\big(1 \pm \sqrt{1-4h}\big)$; flow to the right between them, to the left outside.
- $h = \frac{1}{4}$: one half-stable fixed point at $x = \frac{1}{2}$.
- $h > \frac{1}{4}$: no fixed points; $\dot{x} < 0$ everywhere.

(c) At $h_c = \frac{1}{4}$ a **saddle-node** bifurcation occurs ($f_h = -1 \neq 0$, $f_{xx} = -2 \neq 0$) at $x^* = \frac{1}{2}$.

(d) For $h < h_c$: x_+ is stable (a sustainable harvested population) and x_- is unstable (a threshold). If the population starts above x_- it settles at x_+; if overfishing pushes it below x_- it collapses. For $h > h_c$: there are no fixed points and the population declines to zero (in fact to negative values, which is the unphysical feature of this model): the fishery collapses whatever the initial population. Biologically, $h_c = \frac{1}{4}$ means that the maximum sustainable harvest rate is $H_c = rK/4$. Harvesting above this rate cannot be sustained.

Exercise 3.7.4 (Improved model of a fishery) $\dot{N} = rN(1 - N/K) - H\dfrac{N}{A+N}$.

Solution. **(a)** The catch rate $HN/(A+N)$ saturates at H for $N \gg A$ and equals $H/2$ when $N = A$. So A is the population level at which the fishing rate is half-maximal; it measures how hard the fish are to find (small A: effective fishing even when stocks are low).

(b) Let $x = N/K$, $\tau = rt$, $a = A/K$, $h = H/(rK)$. Then $H\dfrac{N}{A+N} = H\dfrac{x}{a+x}$ and

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = x(1-x) - h\frac{x}{a+x}.$$

(c) $\dot{x} = x\,g(x)$ with $g(x) = (1-x) - \dfrac{h}{a+x}$. Fixed points: $x = 0$ for all parameters, and for $x \neq 0$ the roots of $(1-x)(a+x) = h$, i.e. $x^2 - (1-a)x - (a-h) = 0$:

$$x_\pm = \frac{(1-a) \pm \sqrt{(1+a)^2 - 4h}}{2}.$$

These are real iff $h \leq (1+a)^2/4$; their sum is $1-a$ and their product is $h-a$. Stability: $f'(0) = 1 - h/a$, and at $x_\pm$ (using $h/(a+x) = 1-x$),

$$f'(x_\pm) = x\,g'(x) = x\,\frac{1-a-2x}{a+x} = \mp\frac{x\sqrt{(1+a)^2-4h}}{a+x},$$

so for positive $x_\pm$: x_+ is stable and x_- is unstable. Counting fixed points with $x \geq 0$:

- $h < a$: product < 0, so exactly one positive root x_+. *Two* fixed points: 0 (unstable, $f'(0) > 0$) and x_+ (stable).
- $a < h < (1+a)^2/4$ and $a < 1$: product > 0, sum $= 1-a > 0$: both roots positive. *Three* fixed points: 0 (stable), x_- (unstable), x_+ (stable).
- $h > (1+a)^2/4$, or $h > a$ with $a \geq 1$ (roots negative or complex): *one* fixed point, 0, stable.

(d) Near $x = 0$: expand $\dfrac{hx}{a+x} = \dfrac{h}{a}x - \dfrac{h}{a^2}x^2 + \dots$ to get

$$\dot{x} = \Big(1 - \frac{h}{a}\Big)x + \Big(\frac{h}{a^2} - 1\Big)x^2 + O(x^3).$$

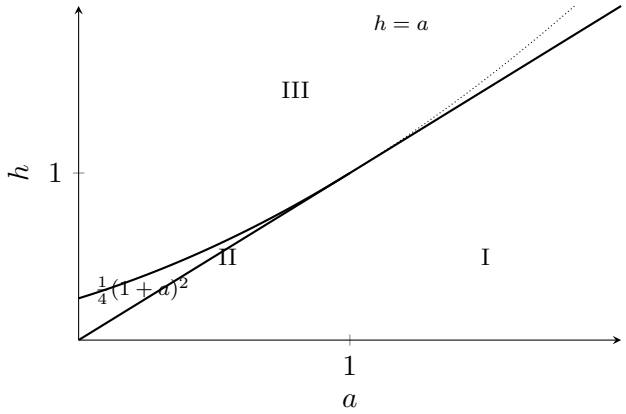
At $h = a$ the linear coefficient vanishes and the quadratic coefficient is $(1-a)/a \neq 0$ for $a \neq 1$. So at $h = a$ the fixed points 0 and the other root exchange stability: a **transcritical** bifurcation.

(e) For $h = \frac{1}{4}(a+1)^2$ the two roots $x_\pm$ coincide at $x = \frac{1}{2}(1-a)$. This is a **saddle-node** bifurcation, and it occurs at positive x (hence is biologically relevant) only for $a < a_c = 1$. At $a = a_c = 1$ the saddle-node curve touches the transcritical line $h = a$ at $(a, h) = (1, 1)$, where $\dot{x} = -x^3/(1+x)$: a codimension-2 (cusp-like) point. For $a > 1$ the root coalescence happens at $x < 0$ and is unphysical.

(f) Stability diagram in the (a, h) plane: the line $h = a$ (transcritical) and the parabola $h = (1+a)^2/4$ for $a < 1$ (saddle-node), tangent at $(1, 1)$.

- Region I ($h < a$): 0 unstable, x_+ stable; the fishery persists at x_+.
- Region II ($a < h < (1+a)^2/4$, $a < 1$): bistable: 0 and x_+ stable, x_- unstable.
- Region III (the rest): only the extinct state 0, stable.

Hysteresis occurs in region II ($a < 1$): increasing h across the parabola from II to III causes an abrupt collapse to $x = 0$ (saddle-node), but then decreasing h back below the parabola does *not* recover the fishery, since 0 is stable throughout region II; recovery requires $h < a$ (crossing the transcritical line into region I). For $a > 1$ there is no region II and the collapse at $h = a$ is smooth and reversible.



(In region II, $a < h < (1+a)^2/4$ only exists for $a < 1$, as drawn; the dotted continuation of the parabola for $a > 1$ is not a bifurcation of positive fixed points.)

Exercise 3.7.5 (A biochemical switch) $\dot{g} = k_1 s_0 - k_2 g + \dfrac{k_3 g^2}{k_4^2 + g^2}$.

Solution. **(a)** Let $g = k_4 x$, $\tau = (k_3/k_4)\,t$. Then

$$k_4 \dot{x} = k_1 s_0 - k_2 k_4 x + k_3 \frac{x^2}{1+x^2} \implies \frac{\mathrm{d}x}{\mathrm{d}\tau} = s - rx + \frac{x^2}{1+x^2},$$

with $s = \dfrac{k_1 s_0}{k_3} \geq 0$ and $r = \dfrac{k_2 k_4}{k_3} > 0$.

(b) For $s = 0$: $\dot{x} = x\big(-r + \dfrac{x}{1+x^2}\big)$. Nonzero fixed points satisfy $r = \dfrac{x}{1+x^2}$, i.e. $rx^2 - x + r = 0$:

$$x^* = \frac{1 \pm \sqrt{1-4r^2}}{2r},$$

which are two positive fixed points iff $4r^2 < 1$, i.e. $r < r_c = \frac{1}{2}$ (the maximum of $x/(1+x^2)$ is $\frac{1}{2}$ at $x = 1$).

(c) For $s = 0$ and $r < \frac{1}{2}$ the fixed points are: $x = 0$ (stable, $f'(0) = -r$), a middle unstable one, and a larger stable one: bistability. Start at $x = 0$ and slowly increase s. For $s > 0$ the origin is no longer a fixed point, but the lower stable fixed point $x_L(s)$ (initially $\approx s/r$) remains and rises slowly with s: a low expression level. When s reaches the saddle-node value at which x_L collides with the unstable middle fixed point, the lower branch disappears and $g(t)$ jumps to the upper stable branch: the gene is switched on. If s is then decreased slowly back to 0, the upper stable fixed point persists (it exists at $s = 0$ for $r < \frac{1}{2}$), so g stays high: the gene does *not* turn off again. The switch is irreversible (hysteresis).

(d) Fixed points satisfy $s = rx - \dfrac{x^2}{1+x^2} =: S(x)$. Saddle-node bifurcations: $S'(x) = 0$, i.e. $r = \dfrac{2x}{(1+x^2)^2}$. Substituting back,

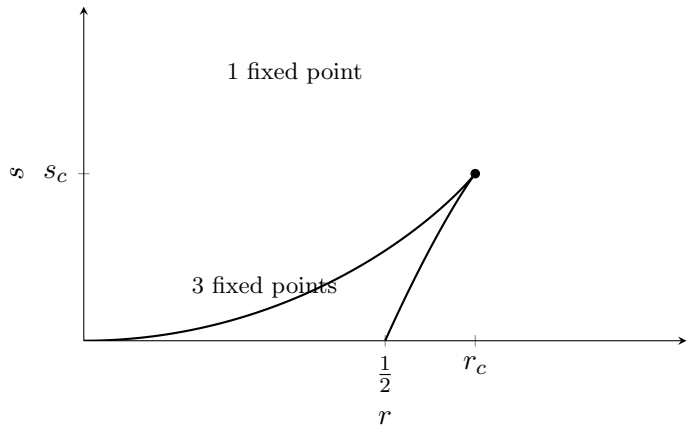
$$\boxed{r(x) = \frac{2x}{(1+x^2)^2}, \qquad s(x) = \frac{x^2(1-x^2)}{(1+x^2)^2}}.$$

Physical values need $s \geq 0$, i.e. $0 < x \leq 1$. At $x = 1$: $(r, s) = (\frac{1}{2}, 0)$, consistent with (b). Cusp: $r'(x) = \dfrac{2(1-3x^2)}{(1+x^2)^3}$ and $s'(x) = \dfrac{2x(1-3x^2)}{(1+x^2)^3}$ both vanish at $x = 1/\sqrt{3}$, giving the cusp point

$$(r_c, s_c) = \Big(\frac{3\sqrt{3}}{8},\ \frac{1}{8}\Big) \approx (0.65,\ 0.125).$$

All the bifurcations on the curves are saddle-nodes; the cusp is the codimension-2 point where they merge.

(e) Stability diagram (parametric plot of the curve for $0 < x \leq 1$). Inside the wedge there are three fixed points; outside, one. For $r < \frac{1}{2}$ the wedge is bounded by the r-axis ($s = 0$) below and by the upper branch ($x < 1/\sqrt{3}$) above; for $\frac{1}{2} < r < r_c$ it lies between the two branches.



Exercise 3.7.6 (Model of an epidemic) $\dot{x} = -kxy$, $\dot{y} = kxy - ly$, $\dot{z} = ly$.

Solution. **(a)** Adding the three equations, $\dot{x} + \dot{y} + \dot{z} = 0$, so $x + y + z = N$ is constant.

(b) Dividing, $\dfrac{\mathrm{d}x}{\mathrm{d}z} = \dfrac{\dot{x}}{\dot{z}} = -\dfrac{kxy}{ly} = -\dfrac{k}{l}\,x$, so $x = x_0 e^{-kz/l}$ (with $z(0) = 0$). Hence

$$x(t) = x_0 \exp\bigl(-kz(t)/l\bigr).$$

(c) $\dot{z} = ly = l(N - x - z) = l\bigl[N - z - x_0 e^{-kz/l}\bigr]$.

(d) Put $u = kz/l$. Then $\dot{u} = \dfrac{k}{l}\dot{z} = k\bigl[N - \frac{l}{k}u - x_0 e^{-u}\bigr] = kN - lu - kx_0 e^{-u}$. With $\tau = kx_0 t$:

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = a - bu - e^{-u}, \qquad a = \frac{N}{x_0}, \quad b = \frac{l}{kx_0}.$$

(e) $N = x_0 + y_0 \geq x_0$ (with $z(0) = 0$), so $a \geq 1$ (strictly > 1 if any sick people are present at $t = 0$); and $b > 0$ since $l, k, x_0 > 0$.

(f) Let $G(u) = a - bu - e^{-u}$. Then $G'(u) = -b + e^{-u}$ and $G'' = -e^{-u} < 0$ (concave), with $G(0) = a - 1 \geq 0$ and $G \to -\infty$ as $u \to \infty$. For $a > 1$ there is therefore exactly one fixed point $u^* > 0$, and it is stable (G crosses from positive to negative). (For $a = 1$ the fixed point $u = 0$ coincides with the origin; it is unstable if $b < 1$ and stable if $b > 1$, with a second positive root in the first case.) So the physical dynamics has a single stable fixed point u^*: the epidemic ends when z reaches $z^* = lu^*/k$.

(g) Since $\dot{z} = ly$, we have $y = \dot{z}/l \propto \dot{u}$. So y and $\dot{z}$ (and hence $\dot{u}$) are maximal at the same time t_{peak}. At that moment there are more sick people, and a higher daily death rate $\dot{z} = ly$, than at any other time.

(h) $\dfrac{\mathrm{d}}{\mathrm{d}\tau}\dot{u} = \ddot{u} = (e^{-u} - b)\,\dot{u}$. At $u = 0$ this is $(1 - b)\dot{u}(0)$, and $\dot{u}(0) = a - 1 > 0$. So if $b < 1$, $\ddot{u}(0) > 0$: $\dot{u}$ is increasing at $t = 0$. It reaches its maximum when $\ddot{u} = 0$, i.e. $e^{-u} = b$, at $u_m = -\ln b > 0$. Since $G(u_m) > 0$ is the global maximum of G and $G(u^*) = 0$, we have $u^* > u_m$, and $u(\tau)$ increases continuously from 0 toward u^*; so it passes u_m at a finite time $t_{\text{peak}} > 0$. Afterwards $e^{-u} < b$ so $\ddot{u} < 0$ and $\dot{u} \to 0$ as $u \to u^*$: it eventually decreases to 0. Things get worse before they get better: an epidemic.

(i) If $b > 1$ then $e^{-u} \leq 1 < b$ for all $u \geq 0$, so $\ddot{u} = (e^{-u} - b)\dot{u} < 0$ for all τ (since $\dot{u} > 0$): $\dot{u}$ is maximal at $t = 0$, so $t_{\text{peak}} = 0$: no epidemic.

(j) The threshold $b = 1$ means $kx_0 = l$. Each sick person transmits to healthy people at an initial rate kx_0 and dies (is removed) at rate l; the ratio $kx_0/l = 1/b$ is the average number of new infections caused by one sick person before dying. The epidemic occurs iff this "reproductive number" exceeds 1, i.e. each sick person infects more than one other.

(k) For AIDS the model should be revised at least in the following ways. (i) Transmission via random mixing at a rate kxy is unrealistic: contacts are structured (sexual networks, risk

groups, needle sharing), so a contact-heterogeneity or multi-group model is needed. (ii) Infected people are infectious for years before symptoms and death, so a constant removal rate l is wrong: one needs additional classes (infected-asymptomatic, symptomatic) and a duration-dependent infectiousness. (iii) Births, immigration and non-AIDS deaths are not negligible on the time scale of the epidemic (so N is not constant). (iv) Behaviour changes with awareness, and treatment prolongs life and reduces infectiousness.

Exercise 3.7.7 (Hysteretic activation) $\dot{p} = \alpha + \dfrac{\beta p^n}{K^n + p^n} - \delta p$, $n \gg 1$.

Solution. **(a)** The function $g(p) = \beta p^n/(K^n + p^n)$ is a sigmoid that sharpens as n grows: for $n \gg 1$ it is ≈ 0 for $p < K$ and $\approx \beta$ for $p > K$, with $g(K) = \beta/2$. As $n \to \infty$ it approaches a **step function** of height β at $p = K$.

(b) Write $\dot{p} = g(p) - h(p)$ with the line $h(p) = \delta p - \alpha$. Fixed points are intersections of the step g with the line h:

- on $p < K$ ($g = 0$): $p = \alpha/\delta$, valid iff $\delta K - \alpha > 0$;
- on $p > K$ ($g = \beta$): $p = (\alpha + \beta)/\delta$, valid iff $\delta K - \alpha < \beta$.

Where $g > h$ the flow goes to the right, otherwise to the left. The three cases:

(i) $\delta K - \alpha > \beta$ (line above the step at $p = K$): only $p^* = \alpha/\delta < K$; the flow goes toward it from both sides. *One stable fixed point (low).*

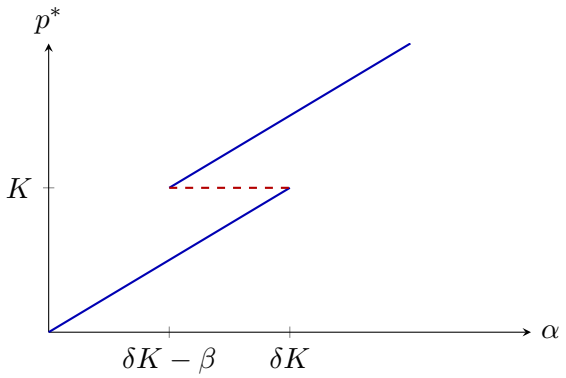
(ii) $\delta K - \alpha = \beta/2$ (line through the middle of the step): $p^* = \alpha/\delta < K$ (stable), $p^* \approx K$ (unstable, at the step), and $p^* = (\alpha + \beta)/\delta > K$ (stable). *Bistable.*

(iii) $\delta K - \alpha < 0$ (line below 0 at $p = K$): only $p^* = (\alpha + \beta)/\delta > K$; stable. *One stable fixed point (high).*

(c) Assume $\delta K > \beta$. In the $n \to \infty$ limit, the fixed points as functions of α are:

- low branch $p^* = \alpha/\delta$, stable, for $0 \le \alpha < \delta K$ (it ends at $p = K$);
- high branch $p^* = (\alpha + \beta)/\delta$, stable, for $\alpha > \delta K - \beta$ (it starts at $p = K$);
- unstable branch $p^* \approx K$ for $\delta K - \beta < \alpha < \delta K$ (joining the two).

For large finite n the vertical unstable segment becomes a slightly tilted curve, and the endpoints are the saddle-node bifurcations at $\alpha_1 \approx \delta K - \beta$ and $\alpha_2 \approx \delta K$ (the S-shaped, bistable window). Diagram (with $\delta = K = 1$, $\beta = 0.5$):



(d) Start at $\alpha = 0$ in the low state, with $p^* = \alpha/\delta$. As α is increased very slowly, p follows the low branch up to $\alpha = \delta K$, where the low branch ends in a saddle-node; p then jumps up to the high branch $p^* = (\alpha + \beta)/\delta$ and keeps rising with α. When α is slowly decreased again, p stays on the high branch (still stable) as α crosses δK, and only falls back to the low branch when α reaches $\delta K - \beta$, where the high branch ends. So the upward jump (at $\alpha = \delta K$) and the downward jump (at $\alpha = \delta K - \beta$) occur at different values of α: a hysteresis loop. Consequently a transient pulse of α above δK switches the protein to the high state, and it remains there for as long as α stays above $\delta K - \beta$: the system "remembers" the pulse.

Exercise 3.7.8 (Irreversible response to a transient stimulus) $\dot{A}_p = k_p S A + \beta \dfrac{A_p^n}{K^n + A_p^n} - k_d A_p$, $A = A_T - A_p$, $n \gg 1$, $K = A_T/2$, $\beta = k_d A_T$.

Solution. **(a)** Substitute $A = A_T - A_p$ and the dimensionless variables $x = A_p/K$, $\tau = k_d t$, $s = k_p S/k_d$, $b = \beta/(k_d K)$. Since $A_T = 2K$, $A = K(2 - x)$, and dividing by $k_d K$:

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = s(2 - x) + b\,\frac{x^n}{1 + x^n} - x.$$

In the special case $\beta = k_d A_T = 2k_d K$ we get $b = 2$.

For $n \gg 1$ the Hill term is a step: $x^n/(1 + x^n) \to 0$ for $x < 1$ and $\to 1$ for $x > 1$ (value $\frac{1}{2}$ at $x = 1$).

(b) $s = 0$, $b = 2$: $\dot{x} = 2\,\Theta(x - 1) - x$ (with the step Θ). Fixed points: $x = 0$ (on $x < 1$ the equation is $\dot{x} = -x$), $x = 2$ (on $x > 1$, $\dot{x} = 2 - x$), and $x = 1$, where $2 \cdot \frac{1}{2} = 1 = x$. The flow: $\dot{x} < 0$ for $0 < x < 1$ (toward 0), $\dot{x} > 0$ for $1 < x < 2$ (toward 2), $\dot{x} < 0$ for $x > 2$. So 0 and 2 are stable and $x = 1$ is unstable: a bistable phase portrait,

$$0 \;\longleftarrow\; 1 \;\longrightarrow\; 2 \;\longleftarrow\; \cdots$$

(arrows point toward 0 on $(0, 1)$, toward 2 on $(1, \infty)$).

(c) For $s > 0$ ($b = 2$): $\dot{x} = 2s + 2\Theta(x - 1) - (1 + s)x$.

- Low fixed point: $2s = (1 + s)x$, $x_L = \dfrac{2s}{1 + s}$, valid only if $x_L < 1 \iff s < 1$. Stable.
- High fixed point: $2s + 2 = (1 + s)x$, so $x = 2$ for every s; stable.
- At the step: just left of $x = 1$, $\dot{x} = s - 1$; just right, $\dot{x} = s + 1 > 0$. For $s < 1$ there is a sign change from $-$ to $+$ across $x \approx 1$: an unstable fixed point near $x = 1$ (for finite large n, at $x \approx 1 - \frac{1}{n} \ln \frac{1+s}{1-s}$). For $s > 1$, $\dot{x} > 0$ on both sides: no fixed point there.

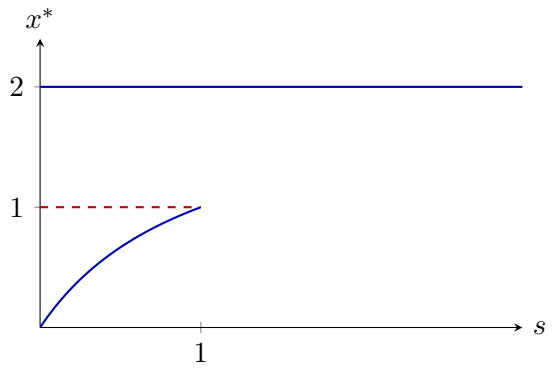
Hence three qualitatively different phase portraits:

(i) $0 < s < 1$: three fixed points: $x_L = 2s/(1 + s)$ (stable), $x \approx 1$ (unstable), $x = 2$ (stable).

(ii) $s = 1$: x_L merges with the unstable point at $x = 1$ (a saddle-node): two fixed points, $x = 1$ half-stable and $x = 2$ stable.

(iii) $s > 1$: a single fixed point $x = 2$, stable; every trajectory flows to it.

(d) Bifurcation diagram of x^* vs. s: the low branch $x = 2s/(1 + s)$ (stable, rising from 0 to 1 as s goes from 0 to 1); the unstable branch $x \approx 1$ for $0 \le s \le 1$; the high branch $x = 2$ (stable, all $s \ge 0$). A saddle-node bifurcation occurs at $s = 1$, $x = 1$.



(e) Begin with no phosphorylated protein and no stimulus: $x = 0$, $s = 0$. Increase s slowly: x follows $x_L = 2s/(1 + s)$ until $s = 1$, where the low branch disappears and x jumps to $x = 2$ (activation). Now decrease s back to 0: $x = 2$ is a stable fixed point for *every* $s \ge 0$ (including $s = 0$), so x stays at 2. The protein stays activated after the stimulus is removed: the response is irreversible.

(f) Now take $\beta \ll k_d A_T$, i.e. $b = \beta/(k_d K) \ll 2$ (so $b < 1$). For general b, $\dot{x} = s(2 - x) + b\,\Theta(x - 1) - x$, with

- low fixed point $x_L = \dfrac{2s}{1+s}$, valid iff $s < 1$;
- high fixed point $x_H = \dfrac{2s+b}{1+s}$, valid iff $x_H > 1 \iff s > 1-b$;
- an unstable fixed point near $x = 1$ when both exist, i.e. for $1 - b < s < 1$.

So the bistable window is $1-b < s < 1$, of width b. Sweeping s up from 0: the system jumps to the high branch at $s = 1$. Sweeping s back down: the high branch persists until $s = 1-b$, where it ends in a saddle-node, and the system drops back to the low branch; as $s \to 0$, $x \to 0$. For $b < 1$ the high state does not exist at $s = 0$ (since $1 - b > 0$), so the activation is **reversible**, with a (narrow) hysteresis loop of width b that vanishes as $\beta \to 0$. Irreversibility requires $b \geq 1$ (so that $1 - b \leq 0$ and the high state survives at $s = 0$), as with $b = 2$ above.
